What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs

This version: 30 October 2026

Abstract

Can market prices tell apart groups of traders who react to past prices at different speeds? In a linear market in which groups follow prices with exponentially fading memory and their combined orders move the price, prices reveal only one combined summary of the groups. Groups with different memories can be told apart, but the cost rises steeply as their memories approach: for K unknown groups whose memories are $\Delta\eta$ apart, the number of controlled experiments grows as $(\Delta\eta)^{-(4K-2)}$, with matching bounds on suitably rich classes, and watching without trading needs at least as many observations. For a known pair, the trading capital needed rises sixteen-fold each time their difference is halved. Records that report each group's flow separately keep the problem well conditioned, and a stylised supervisory decision shows when that matters. In pre-specified tests, published trader-class positions in eight CFTC futures markets, and Binance class data in 29 of 29 new markets, pin down a position persistence that prices leave open, and the predicted rates hold with real Bitcoin price noise where it can be used directly. We do not show that this persistence is learning memory, and several pre-specified tests failed, including a test of what prices do reveal and a test of CFTC class flows.

Keywords: algorithmic trading, identification, trader heterogeneity, market microstructure, system identification, minimax sample complexity, value of disclosure, audit trails.
JEL: C14, C32, C58, D83, G14, G18.

Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix I maps each table,

figure and number to its source. "Registered" or "pre-specified" in this paper means that the authors wrote the specification and committed it to the public repository before opening the test data; this is the authors' declaration, not independent preregistration. For the October 2026 follow-up tests (Appendix F and Section 6.5) each specification reached the public repository in the same push as its results, so the order rests on the commit history alone. For the real-noise test, the class-flow test and the CFTC robustness checks (Sections 5.5 and 6.4), specifications and code were pushed in a separate commit (`84af69f`, 03:43 UTC on 31 October 2026) before the first of them ran (03:56 UTC). The proofs have been checked by the authors and against numerical computations, and they have not yet been independently reviewed; the numerical checks reproduce results with the same code unless stated otherwise. Lean 4 with Mathlib machine-checks, using only Lean's standard axioms, two of the main results: Theorem 3.1(i) for the model itself (for known a and $\kappa \neq 0$, two economies have the same price response if and only if they have the same merged measure, and with distinct nonzero rates the transfer function has exactly $1+K$ poles; the classical step from pole count to minimal state dimension is not formalized), and the lower bound of Theorem 5.4 on a finite observation horizon, from the moment gap through the closed loop and the Gaussian divergence to the testing step. The check holds for every K with no added assumption: in place of the Gauss and Gauss–Radau rules it uses another moment-matched pair, binomial weights on $2K$ equally spaced nodes, which gives the same order; its constant depends on the number of observed periods per experiment. Lean also checks the reset-experiment form of the tagged-flow result: class-tagged records recover every group's weight with an error bound that does not depend on the memory rates. The upper bound (Theorem 5.5), the general-memory and passive-data theorems and the central limit step of Proposition 6.1 are not formally verified; Supplement C.7 reports numerical stress tests of them. The results whose proofs most need an independent check are Theorems 5.5, 5.7, 5.9, D.2 and D.4. Numerical constants such as the capital bounds are computed at a declared parameter set and are not separate theorems.

1 Introduction

Markets are increasingly run by algorithms. Trend-following funds, market-making programs, execution algorithms and, more recently, AI trading agents all read recent prices, decide what to trade, and by trading move the prices they read. When something goes wrong, a sharp sell-off, a sudden loss of liquidity, a crowded trade unwinding, the first questions supervisors and risk managers ask are who was trading and how. Which group of traders drove the move? How quickly do they react, and how long do they remember what prices did? Could the same group do it again?

This paper asks how much of that can be learned from what everyone can see, prices and anonymous order flow, and, when public data are not enough, what it costs to find out in other ways.

The question matters now. Trading is passing to learning algorithms and AI agents (Colliard et al., 2022; Dou et al., 2025), and regulators have begun to require firms to say which algorithm sent each order. Under MiFID II, EU trading venues must keep order records that identify the person or algorithm responsible for each investment and execution decision (European Commission, 2017; ESMA, 2016). The US Consolidated Audit Trail (CAT) is designed to record each order in US-listed stocks and options, and the account behind it, across exchanges and broker-dealers; its revised 2026 budget, published in March 2026, is about $147 million (SEC, 2012; CAT LLC, 2026). Such systems serve many purposes, among them enforcement, rebuilding market events after the fact, and surveillance. One question they raise, and the one we study, is how much of the information about who drives price moves could instead be learned from public data, and at what cost.

The short answer. Prices tell an observer less than one might hope, and sharpening what they do tell is expensive.

- Prices reveal one combined summary of all the trader groups, not the groups themselves. Two groups that react to prices with the same memory look exactly like one

larger group, however much price data is collected (Theorem 3.1).

- Groups with different memories can be told apart in principle, but the data needed grow very fast as their memories get closer. For two unknown groups, halving the difference between their memories multiplies the data needed by 64; for three groups, by 1,024 (Theorems 5.4 and 5.5).
- Records that report each class's flow separately remove this problem. Each group then becomes an ordinary estimation problem that stays well conditioned however similar the groups are (Proposition 6.1).

We keep three statements apart throughout: what cannot be learned at all, what can be learned but only at high cost, and where estimates become unstable.

Why prices hide the groups. The reason is the feedback loop. Algorithms trade on recent prices, their orders move prices, and those prices feed back into the algorithms. We model each group as tracking the price with a fading memory: recent prices count most, older prices count less, and one number, the group's memory rate, sets how fast the past fades. A desk behaves exactly this way if it uses a Kalman filter or a regularized online learning rule (Lemma 2.1). Because every group reaches the price only through the sum of all their trades, the price records the total and loses the parts. Think of several people pushing the same door: from the door's movement one can learn the total push and how it builds and fades, but two people who push in the same rhythm look like one stronger person.

Three ways to learn more, and what each costs. An observer that wants to know more, whether a supervisor, an exchange's surveillance team or a large trading desk, has three options (Section 2.2).

(a) *Wait and watch prices.* This costs time. In a stationary Gaussian market observed without error, a passive observer needs at least as many observations, in order of magnitude, as an active one needs experiments (Theorem 5.9).

(b) *Trade against the groups* and watch how the price responds, much as one taps a wall to find a hidden beam. Every test trade moves the price against the trader, so learning this way costs money. To tell one group from two groups whose memory rates differ by 2δ, the quantity traded grows like $1/\delta^2$ and the capital at risk like $1/\delta^4$: halve the difference and the capital needed rises sixteen-fold (Theorems 4.1 and 4.3). If memory is a general mixture of fading memories rather than a few exponentials (with the slowest bounded away from a random walk), accuracy improves only with the logarithm of the data, so halving the error squares the data needed (Theorem 5.7).

(c) *Acquire labelled records*, such as audit-trail data or class-level position reports. This costs whatever the reporting system costs, but it removes the explosion in data: labels save a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy, where K is the number of groups and $\Delta\eta$ the spacing of their memory rates (Theorem 6.2).

Proposition 6.3 says when each option is the cheapest.

How large are the costs? A hypothetical illustration. To give a sense of scale we price the trading option in a hypothetical benchmark. The model parameters are illustrative; only the size of the measurement noise is estimated, from Bitcoin data. An outside participant that re-learns two unknown algorithms each week with test trades would lose at least \$15 million to \$356 million a year per market; with three unknown algorithms it would lose more than \$40 million a year per market even when their memories are far apart (Section 6.2). These are lower bounds for one stated way of learning, not estimates of what any regulator spends: regulators do not trade to learn. They are best read as the price of the information that labelled records deliver without any trading.

Why this matters for finance. The results point in a constructive direction for several groups of readers.

- *Supervisors and policymakers* get a way to decide what data to require. If the goal

is only to watch overall price risk, public prices are enough, because the combined price response is learned at the ordinary rate (Remark 5.3). Labelled records earn their cost when supervision must be targeted, for example when one algorithm must be paused while the others keep supplying liquidity, and when the reporting system covers enough markets to share its fixed cost (Proposition 6.3). In our stylised cost model this illustrates a possible economy of scale for market-wide reporting; it does not explain the legal design of regimes such as CAT. For the narrow task of attributing behaviour to classes, a persistent pseudonymous label is enough, which bears on how much personal data such a task needs but not on what enforcement needs (Section 7). Proposition 6.4 gives a stylised decision in which this matters: the size of a market-wide measure depends only on the combined response, while deciding whether halting one group is enough needs that group's own intensity and inherits the full cost of separating the groups.

- *Exchanges and trading venues*, which already see account-level flow, learn which questions about their participants public prices cannot answer, and why publishing class-level positioning, as some crypto venues already do, adds information that prices do not contain.
- *Risk managers and trading desks* get a realistic limit on what can be inferred about other traders or about crowding from market data, and a warning that common feedback measures computed from public data mostly reflect filtering and timing rather than true feedback (Appendix H).
- *Researchers* who estimate models with several trader types get one reason why the parameters that separate the types are often imprecise, and a number that says how precise they can be before any estimation (Section 1.1).

What the data show. Two real reporting systems let us check the main prediction, that class records pin down what prices leave open. The U.S. Commodity Futures Trading

Commission collects positions from large futures traders and publishes a weekly snapshot of them, aggregated by trader category. In a pre-specified test in eight futures markets (CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, crude oil and natural gas), these class positions pin down the persistence of the designated speculative class's positions to a range 9 to 71 times narrower than weekly prices allow, while prices alone leave it almost anywhere in the admissible range (Section 6.4). On Binance, in a pre-specified test on 30 markets never used before that matches the memory horizon of the two routes, the class data give the narrower interval in 29 of 29 evaluable markets, by a median factor of 7.3 (Section 6.5).

Robustness, a new failure and real noise. Rerunning the CFTC test with CME futures prices, without contract-roll weeks or with longer bootstrap blocks leaves its primary result unchanged. Read as weekly flows rather than positions, the same reports failed a second pre-specified test: they pin down how each class's flow responds to prices, but not how persistent that flow is. A third pre-specified test gives synthetic groups real Bitcoin price noise instead of Gaussian noise and finds the predicted rates, a slope of -6.01 against -6 for two groups and -9.81 against -10 for three, while class-tagged requirements stay flat; at large sample sizes it uses a normal approximation to the real noise (Section 5.5). These comparisons are descriptive: a position series reveals its own persistence directly, while the price route must infer it. Several predictions failed, and Table 1 lists them next to the ones that passed. The data therefore show what prices cannot pin down. They do not show that this persistence is the learning memory of the model.

Two supplementary results. Neither is needed for the main results. Markets with the same eigenvalues, the same one-step volatility and the same long-run price variance can still have very different worst-case short-run losses, so watching eigenvalues or volatility cannot replace knowing who trades (Theorem G.1). And standard feedback measures computed from public data mostly reflect filtering and clock alignment: two fee changes that moved

Table 1: What the empirical tests found: passes and failures

Test	Prediction	Result
CFTC, 8 markets, H1	Class positions pin down a persistence that weekly prices leave open	Pass, 8 of 8
CFTC, H2	The designated speculative class trend-follows	Fail, 4 of 8 (three markets have the opposite sign)
CFTC, H3	That class has the less persistent positions	Pass, 6 of 8 (two indistinguishable)
CFTC flows (weekly changes), H1	Class flows pin down their own persistence (95% interval narrower than 0.5)	Fail, 3 of 8 (6 required); flow response nonzero in 6 of 8
Real-noise injection, $K = 2$	Price-only requirement grows as $(\Delta\eta)^{-6}$; tagged requirement stays bounded	Pass: slope -6.01 (large-N points use a normal approximation); tagged max/min 2.8 (≤ 8 required)
Same test, $K = 3$ (secondary)	Price-only requirement grows as $(\Delta\eta)^{-10}$	Holds: slope -9.81
Binance, 30 fresh markets, (i)–(ii)	Class data restrict persistence; price-only intervals wide	Pass, 30 of 30 (one-hour price kernel)
Binance, same test, (iii)	Prices pin down the combined response	Fail, 5 of 30; the test failed its 80% criterion
Binance, 30 new markets	Horizon-matched: class interval narrower	Pass, 29 of 29 evaluable (median factor 7.3)
Binance, same test, secondary	Class persistence is a stable memory	Not supported: close to a random walk after bias correction
Binance, 105-day follow-up	Combined response far cheaper to learn than the separate rates	Fail, 10 of 30 (23 required)
Binance, 58-day follow-up	Tagging advantage grows with the sample; parametric shrinkage	Pass, 22 of 29; fail, 20 of 29 (22 required)
Binance, return forecasts	Class memory states improve 30-minute return forecasts	Fail ($\Delta R^2 = -1.09$ points, inconclusive)
Binance, state tracking	Class data track class positions better than prices	Pass, but explained by persistence of the target

trading volume by factors of 2.6 and 4.9 left measured feedback unchanged (Appendix H).

Contribution. A large literature asks how transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006). We ask a different question that market supervision now faces as trading passes to algorithms: what can an observer learn about *who* moves prices, and what is it worth to be told? The paper makes three contributions. First, it gives an identification result for supervision: a closed-loop price process reveals only the combined distribution of the trader groups' memories and weights, and nothing finer (Theorem 3.1), so questions about individual groups cannot be answered from public prices however much data accumulates. Second, it prices what it takes to go further, in the units a market participant or regulator faces, the number of experiments or observations, the quantity traded and the capital at risk, with matching upper and lower bounds where we can prove them (Sections 4 and 5); the cost explodes as $(\Delta\eta)^{-(4K-2)}$ when groups react over similar horizons, which turns the economic value of attribution into a number. Third, it values disclosure in the same units and shows when it changes a decision: class-tagged records keep the problem well conditioned (Proposition 6.1), sizing a market-wide limit needs only public prices, and deciding whether halting one group is enough needs the records (Proposition 6.4); published CFTC and Binance class positions pin down what prices leave open, with the failed predictions reported next to the passes (Section 6). The mathematical tools, moment matching, two-point testing and Prony-type estimation, are known from mixture estimation and signal processing; what is new is their closed-loop, passive and financial form (Section 1.1). Two of the central results, the identification theorem and the lower bound, are machine-checked in the Lean proof assistant, which is still rare in economics and finance.

How to read the paper. Every main result is stated formally and then explained in a short paragraph headed "In words", which says what it means for a trader or a supervisor; proofs are in the online supplement. Section 4.5 works through two desks that use Kalman

filters, in numbers, and Table 5 maps each question to its answer. Readers who want the finance first can read this introduction, Section 2.2, the worked example and Section 6, and take the statistical results as given. Table 2 explains the main symbols.

Table 2: The main symbols in plain words

Symbol	Meaning
K	Number of trader groups ("cohorts")
η_i	Group i's learning rate: the weight it puts on the newest price. Large η_i means a short memory and fast reactions; small η_i means a long memory
$r_i = 1 - \eta_i$	Group i's memory rate: the share of last period's view that survives
q_i	Group i's trading intensity: how strongly it trades on its view. Positive q_i means it buys when the price is above value, as trend followers do
$\Delta\eta$, 2δ	How far apart the groups' learning rates are ($\Delta\eta$ for unknown groups, 2δ for a known pair)
a, κ	How persistent pricing errors are, and how far one unit of net order flow moves the price
M (merged measure)	The combined summary of all groups that prices reveal: where the memories are and how much trading sits at each
N	Number of experiments (test trades from a fresh start) or observations
Par units	The model's own unit of quantity and money, before conversion to USDT (Table 27)

What we do not claim. The identification results hold for the stated linear model and for the stated observation contracts, that is, for what the observer is assumed to see. The model treats price impact and the persistence of pricing errors as known primitives rather than solving a dealer equilibrium, and it assumes the observer knows the fundamental value; real data meet neither assumption. They are not general limits on surveillance with participant-level records, which can be far more informative (Propositions 4.4 and 6.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-specified real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Section 6.5). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 6.1, not

an instance of it. We hold the algorithms' memories and intensities fixed across disclosure regimes; disclosure that reveals positions may lead informed desks to trade less or to split orders across accounts, which we do not model.

1.1 Related literature

Algorithms, AI and supervision. Recent work in finance asks how learning algorithms and AI agents behave in markets: reinforcement-learning market makers and informed traders can sustain wide spreads or collusive outcomes (Colliard et al., 2022; Dou et al., 2025), prices formed against Q-learners have distinctive dynamics (Banerjee and Szydlowski, 2026), and algorithmic markets can be manipulated (Tremacoldi-Rossi, 2025). Traders who differ in horizon or speed are studied in equilibrium by Cespa and Vives (2015) and Biais et al. (2015), and Crouzet et al. (2020) model investors who trade at different frequencies. These papers model or simulate the algorithms. We take the supervisor's side and ask what can be learned about them from market data, and what disclosure is worth. A related literature measures the cost of regulation and of regulatory reporting (Trebbi et al., 2024); our dollar comparison with the Consolidated Audit Trail is a statement about the benefit side: what the audit trail buys that prices cannot.

Price formation and impact. Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Gatheral (2010) shows that absence of dynamic arbitrage requires a positive-definite decay kernel, Alfonsi et al. (2012) that excluding transaction-triggered manipulation requires convex, nonincreasing decay, and Alfonsi et al. (2016) extend positive definiteness to several assets. Positive definiteness is the property Lemma 4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order

flow, which is the main confounder in Appendix H. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

Algorithmic trading and feedback. Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

Learning algorithms and what can be inferred about them. Empirical game-theoretic analysis treats the choice among simulated trading strategies as a game (Wellman, 2006). Empirically, algorithmic trading has been found to improve liquidity and price discovery on average (Hendershott et al., 2011; Brogaard et al., 2014). These studies simulate the algorithms or observe them through regulatory data; we ask what an outside observer can infer about them from market data alone, and answer in samples and in capital. Our stance is that of partial identification (Manski, 2003): when the structure is not identified, characterize what the data do determine, here the merged measure and, through it, the intensity-weighted distribution of the desks' information quality (Proposition 4.6).

Control without identification. We do not claim that acting on a system without identifying it is new. Moallemi et al. (2008) study strategic execution against an observer who learns from prices, which bears directly on our assumption that cohorts do not react to being observed, and Park and Van Roy (2015) study execution that learns price impact while trading, with regret guarantees; our object is different, the latent trader cohorts behind the impact rather than the aggregate impact coefficients. Active identification of linear systems

has matching bounds and optimal experiment design (Wagenmaker and Jamieson, 2020), and task-directed exploration asks what must be learned for a decision (Wagenmaker et al., 2021); our information matrix differs because the unknowns are zeros of a closed-loop response and the input is paid for in trading losses. van Waarde et al. (2020) characterize when control is possible without unique system identification. Our results concern a specific market observation contract and its costs in executed quantity, capital and money.

Closest work. The closest literature is super-resolution and moment-based mixture estimation, which gives rates for recovering clustered point sources and mixing measures from noisy moments (Moitra, 2015; Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our K-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table 3): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 3.1); with a and κ known this extraction is a smooth invertible map, so it decides which method is valid (pole fitting on prices is not) rather than the rate; second, passive price observations face at least the same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing, uniformly in the spacing, because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 5.8 and 5.9); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^\top T_h v$ (Theorems 4.1–D.4); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$ (Proposition 6.1 and Theorem 6.2). Beyond the K-cohort case, we prove matching lower

and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem 5.7) and price the resulting disclosure savings against audit-trail budgets. Several ingredients are known in other settings, and we claim only their closed-loop, passive and financial forms: the δ^{-4} law for telling one component from two closely spaced ones is classical in resolution analysis (Shahram and Milanfar, 2005); logarithmic rates are known for mixtures of exponentials (Rebafka and Roueff, 2015) and for general mixing distributions, by the same moment-matching and polynomial-approximation technique (Wu and Yang, 2020); and labelled samples are known to be far more informative than unlabelled ones in mixture problems (Castelli and Cover, 1996).

Estimating heterogeneous-agent models from prices. A large literature models markets as populations of trader types that switch between forecasting rules (Brock and Hommes, 1998) and estimates such models from price data. Estimates of the parameters that separate the types are often imprecise: Amilon (2008) reports that the estimated model reproduces properties of returns but that its fit is generally poor, Franke and Westerhoff (2012) compare switching mechanisms by the method of simulated moments, and Kukacka and Barunik (2017) find the switching coefficient statistically insignificant by simulated maximum likelihood while the belief parameters are significant. These papers take the identification of the types from prices as given. Our results suggest one reason the type-separating parameters are hard to pin down: around a steady state, prices determine only the merged measure, and separating types whose memories differ by $\Delta\eta$ takes a number of samples of order $(\Delta\eta)^{-(4K-2)}$ for K types (Theorems 3.1 and 5.4). The models these papers estimate are nonlinear, so this is a statement about their linearization, not a claim that their estimates are wrong; it does give a quantity, the separation of the types' memories, that bounds the precision one can expect before estimating.

Closed-loop system identification. Engineering has long studied identification under feedback (Ljung, 1999; Forssell and Ljung, 1999). There the object is the plant, the difficulty

is that feedback correlates the input with the noise, and the remedy is external excitation that makes the experiment informative. Our problem is the reverse in two ways. The object is the feedback law (the traders), not the plant (the price), and the memory rates enter as zeros of the observed transfer function. And external excitation does not remove the difficulty: with a probe the merged measure is identified exactly (Theorem 3.1), so the barrier is not qualitative identifiability but the conditioning of the map from memories to responses, which we measure in samples, quantity and capital. In finance, Leach and Madhavan (1993) show that a market maker can gain by setting prices to learn about demand, so learning through trading has an equilibrium counterpart; our probing observer is a reduced-form version of that idea without the equilibrium. In econometric terms, as $\Delta\eta \to 0$ the cohort parameters pass from strong to weak identification (Andrews and Cheng, 2012), and our rates state that passage in samples.

Inferring trader types from market data. Market microstructure has a tradition of inferring hidden trader populations from public data. The probability of informed trading (Easley et al., 1996) fits a mixture model to daily counts of buys and sells to estimate the share of informed orders, and what it captures has been debated (Duarte and Young, 2009). With regulatory account data, Kirilenko et al. (2017) classify traders by their holding and trading patterns. Our results speak to the case where types differ in how long they remember prices: from prices alone the separation is costly, and account or class records of the kind Kirilenko et al. (2017) use remove most of the cost (Proposition 6.1).

Statistical identification. The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 5 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and Wu and Yang (2020),

Table 3: Comparison with finite-mixture estimation and open-loop identification

Feature	Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020)	Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021)	Closed-loop market identification (this paper)
Where rates live	Component locations / moments of density	Poles of impulse response	Transmission zeros of closed-loop $H(z)$ (Thm 3.1)
Active vs. passive data	i.i.d. draws only	Designed input vs. output spectrum	At least the same $(\Delta\eta)^{-(4K-2)}$ order, uniformly, via the minimum-phase Whittle gap (Thm 5.9)
Economic unit of cost	Sample size N	Signal-to-noise ratio	Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 4.1–D.4)
Effect of class tagging	Observed component labels ($\Theta(1)$)	Separate sensor per mode	Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-2)})$ at fixed accuracy, $\Theta((\Delta\eta)^{-(4K-4)})$ for rates (Thm 6.2)

and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, the passive equivalence, and the value of class records; that the relevant rates are zeros, not poles, of the observable transfer function matters for which estimator is valid, not for the rate.

Non-normal dynamics and supervision. Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem G.1), which motivates the identification-free control of the companion paper. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure

Table 4: Closest results and how this paper differs

Existing result	Observations	Target	Conclusion	Difference from this paper
Mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020)	i.i.d. draws from a K-component mixture	Mixing measure	Rates that slow sharply as components merge; $n^{-1/(4K-2)}$ for Gaussian location mixtures	Same exponent; here the measure is observed through a feedback loop, as zeros of H, actively or passively, and the cost is also priced in quantity and capital
Super-resolution and Prony (Moitra, 2015; Batenkov et al., 2021; Katz et al., 2023)	Noisy Fourier or exponential samples	Point-source locations	Stability governed by Vandermonde conditioning and node separation; Prony attains the min-max bound for clustered exponents	Real nodes extracted from a closed loop; our matching upper bound (Theorem 5.5) adapts this known result
Closed-loop system identification (Ljung, 1999; Forssell and Ljung, 1999)	Input–output data under feedback	The plant	Consistency requires informative (externally excited) experiments	Target is the feedback law (the traders); excitation gives exact identification of M, and the barrier is conditioning, measured in samples and capital
Heterogeneous-agent estimation (Brock and Hommes, 1998; Franke and Westerhoff, 2012; Kukacka and Barunik, 2017)	Prices and returns	Type and switching parameters	Type-separating parameters often imprecise	Bounds that precision by the types' memory separation (in the linearization)
Probability of informed trading (Easley et al., 1996)	Daily buy and sell counts	Share of informed orders	Mixture estimate of trader population	Types that differ in memory; price-only versus class-tagged data
Data informativity (van Waarde et al., 2020)	Finite data, often noiseless	Whether a property is determined	Qualitative conditions	Quantitative minimax costs under noise

regime for Proposition 6.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

1.2 Outline

Section 2 sets out the model and Section 3 the identification boundary. Section 4 turns memory resolution into executed quantity, capital and money for a known pair, and Section 5 treats unknown mixtures, general memory and passive data. Section 6 states what disclosure buys and when it changes a supervisor's decision, and tests it on the CFTC's mandated trader-class reports and on Binance's class ratios. Section 7 draws the implications for supervision and Section 8 concludes. Proofs are in Appendix A; the internet appendix contains account-level capital, near-colliding rates, robustness to learning rules, the spectral blind spot and the measurement of reduced-form feedback, and Appendix I maps every number to its source.

2 Model and motivating examples

The model in words. The market has one price and K groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value,

and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight η_i on the newest price and $1-\eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity q_i. A group with a large η_i has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small η_i has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1-\eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0,1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because cohorts update on

the same period's price, the probe enters every state on impact through

$$b = \kappa\,(1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1 - \eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2 \sigma_\nu^2}}{2\sigma_\nu^2} \in (0, 1). \tag{7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0, 1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1 - \eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0, 1)$ and $q = \alpha H / \eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.*

Remark 2.2 (Why exponential memory). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk i chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1 - \delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \leq \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i} u_{i,t}^2$ and quadratic inventory holding cost

$\frac{1}{2}\gamma_{\text{inv},i} I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i}+\gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},i}+\gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one. This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in the companion paper).

Two further simplifications matter for reading the results: the observer is taken to know the fundamental value p^*, so the pricing error x_t is observed, and the dealer sector is a fixed linear impact κ rather than an equilibrium quote. The paragraph before Section 6.4 states what the data tests lose when neither holds. Assuming that the observer knows p^* gives it more information than any real observer has, so the lower bounds hold a fortiori when p^* is unknown; the upper bounds use it.

The model applies these rules to the pricing error x_t: a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a signal of a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). With $q_i > 0$ such a desk buys after rises, as trend followers do, and nothing in the model says this is profitable; the lemma shows only that the update has the exponential form. In these two illustrative families of algorithms the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio or its regularization. That is the kind of private information a supervisor would like to recover, although real trading algorithms need not belong to either family. Section 3 shows why prices reveal it only in merged form.

Lemma 2.3 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_{\ell} \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_{\ell} \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J

is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

As an illustration of the lemma (not part of its proof), $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.3 matters twice: it makes probing costs positive (Section 4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

2.1 Discussion of the assumptions

Linearity. The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

Price-first timing. We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

Exponential memory. Lemma 2.1 justifies exponential memory for filtering and regularized learning. Neural and reinforcement-learning policies are not exponential filters. Rules whose orders are smooth functions of an exponential-memory state are exponential filters to first order, so the results hold for small probes; Appendix F checks this in stylized simulations and finds where it fails, and how closely production algorithms follow such rules remains an empirical question we do not settle. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of cohorts, for which Theorem 5.7 shows that accuracy improves only logarithmically in the

data.

Algorithms that react to being probed. The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not help separate them, and an algorithm that mimics another group's memory rate only brings the hypotheses closer, so neither lowers the costs we derive. A reaction that differs across cohorts is a new signature, but one the observer must specify and estimate, so it shifts the identification problem rather than solving it. Second, the passive route needs no probe: Theorem 5.9 shows that natural price variation faces a lower bound of the same polynomial order, so an observer who avoids detection by not trading does not escape the barrier; it pays in time instead of money. We do not show that passive data attain that order. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Appendix D trades at most 1.25 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

Known primitives. The resource theorems take κ, a and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid. The upper bounds as stated assume the primitives known; estimating them adds a cost we do not analyse, which we expect to be of lower order because they enter the response at first order. In an equilibrium in which dealers set the impact κ from the order flow they see, as in Kyle (1985), a prober's unexpected trades would also move κ; we do not model this, and expect it to raise the cost of probing rather than lower it.

Observation contracts. Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

2.2 The observer, its options and its costs

The results are stated from the side of one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. Its *target* is attribution, recovering the merged weight and memory rate of each group to a stated accuracy, or deciding between two candidate populations at a stated error. Before it observes, it knows the price persistence a, the impact κ and the law of the observation noise, and, for the K-group results, the number of groups; it does not know their rates or weights. These are the conditions under which the lower bounds are statements about the data rather than about the observer's ignorance; unknown primitives can only enlarge the set of candidate populations.

The observer has three options, and each result prices one of them.

(a) *Wait.* Record stationary prices driven by natural innovations, without trading. The cost is time: the number of observations needed (Theorems 5.8 and 5.9).

(b) *Probe.* Execute a sequence v_t of trades against the groups and record the price response. Probes execute at the average price of a linear book after the uncontrolled state update; the prober bears the execution loss $\frac{1}{2}v^\top T_h v$ plus fees (Lemma 4.2) and must fund margin throughout. The counted resource is the number of independent experiments, each starting from the reset state, or the executed quantity and capital of a single experiment (Theorems 4.1–D.2).

(c) *Acquire disclosure.* Receive each class's flow or position, observed with noise, at an operating cost per supervisory cycle and a compliance cost per market (Proposition 6.1, Theorem 6.2, Proposition 6.3).

Within each period the order is: the probe (if any) executes and the price updates, the groups update their states on the new price, and the observer records the price with noise. The groups do not change their rules in response to being observed; Section 2.1 discusses what happens if they do. Probing experiments start from the reset state. A probe sent into a running market adds the free response of the current state, which by linearity superposes with the forced response; if the state is unknown it acts as a nuisance term, which can only make separation harder, so the lower bounds still hold, while the upper bounds as stated need the reset state or a known one. Independent resets are a benchmark experiment that no outside participant can run in a live market; we use them because they make the information content of the data exact, and the economic comparisons say whenever they rely on them. Regulators do not, in practice, trade to learn; the probing option describes an outside participant, or prices the information a supervisor would otherwise have to buy by trading.

Three money quantities appear and are kept apart: the *capital required*, the equity an account must hold so that its margin constraint is never breached while it probes (Theorem 4.3 and Lemma 4.5); the *round-trip trading loss*, the execution loss of a probe that ends flat, which is what the account actually spends (Theorems D.2, D.3 and D.4); and the *annualised loss* of renewing that probe weekly, used for the comparison with audit-trail budgets (Section 6.2). The lower bounds hold for the stated probe classes, deterministic probes and, in Theorem D.3, non-anticipating sequential probes; they are not lower bounds on information obtained through other channels, which is the point of Section 6. Two rates recur and measure different things: δ^{-4} is the capital (and the average-power horizon) for a *known* symmetric pair whose rates differ by 2δ, while $(\Delta\eta)^{-(4K-2)}$, which is $(\Delta\eta)^{-6}$ for $K = 2$, is the number of experiments for K *unknown* groups with spacing $\Delta\eta$.

3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{8}$$

Write $a' = a + \kappa M(\{0\})$, so that cohorts with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0, 1)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with rates bounded away from 0 and 1. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \leq P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\geq\; \frac{4\sigma_\nu^2(1 - 2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

where $C_{\mathcal{F}}$ depends only on the family.

In words. If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much

trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 5: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, Φ is the standard normal distribution function.

4.1 Quantity

Theorem 4.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0, 1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1 - \epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0, \sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^{\delta}_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1 - S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$*,*

$$D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{10}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e_T^*(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{11}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$*, so it cannot make the error arbitrarily small at fixed* V*. Error at most* $\alpha < 1/2$ *requires* $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ *independently of* T*.*

(iii) The first two taps coincide and $d_2 = -\kappa^2 q \delta^2$*. A single pulse of size*

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{12}$$

and the third observation achieve error exactly α*. The necessary and sufficient quantity order is* δ^{-2}*.*

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$*, the necessary horizon order is* δ^{-4}*, attained by the pulse whenever* $T \ge \max\{3, \lceil A^2/P \rceil\}$*.*

(v) For a fixed monetary price exposure $\lambda \ne 0$*, additive reserve* L *and cost envelope* $c\sum|v_t|$*, the sharp robust quantity allowance under budget* $B > L$ *is* $V_B = (B-L)/(|\lambda|\kappa + c)$*.*

In words. To tell one desk from two desks whose memories differ slightly, the observer must trade against them, and the quantity it must trade grows like $1/\delta^2$: halving the difference quadruples the trading needed. Waiting longer after a fixed trade does not help, because the trade's echo in the price dies out.

The identity in (iii) follows by direct substitution, and code checks it at $\delta \in$

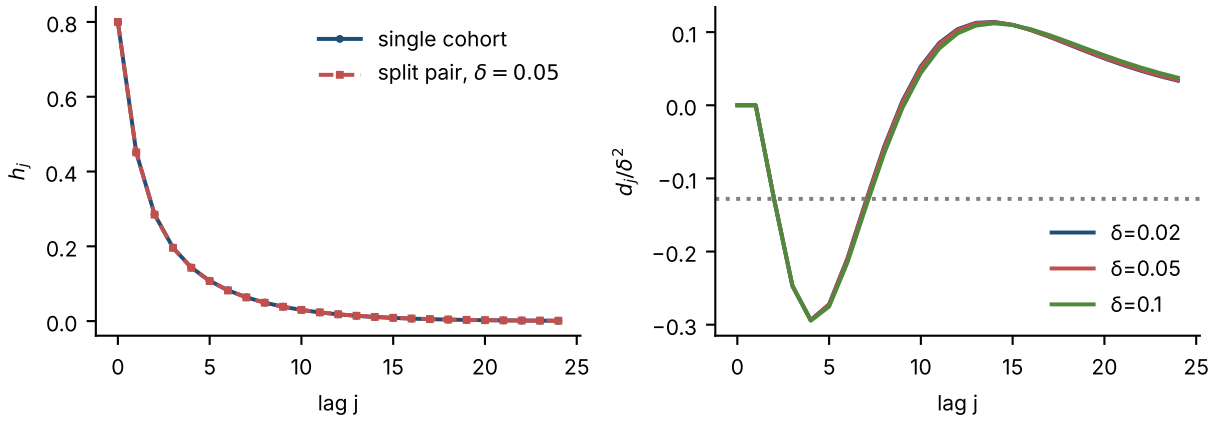


Figure 1: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by δ^2 collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

$\{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 5).

A Monte Carlo check of the exact error. Formula (11) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table 6); simulated errors match the formula within Monte Carlo standard error under both hypotheses.

Figure 1 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma 4.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{13}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

We verify the ledger identity (13) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

4.3 Capital supplied by a counterparty

Theorem 4.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{14}$$

equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

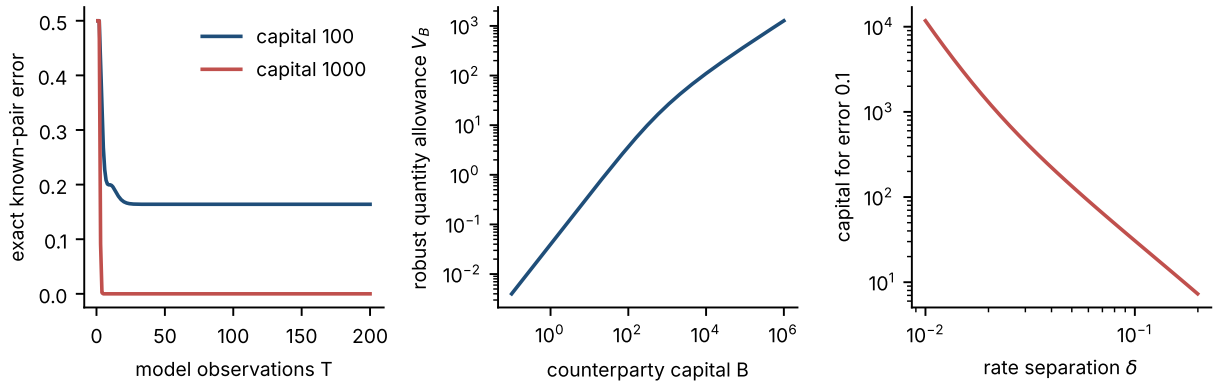


Figure 2: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (14). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

In words. A probe moves the price against the account that makes it, so the margin it needs grows with the square of the probe. Because the probe grows like $1/\delta^2$, the capital grows like $1/\delta^4$: halving the difference multiplies the capital needed by 16.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure 2 illustrates the experiment.

Proposition 4.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse* $0 < A \le V_B$ *distinguishes the known pair with zero error for every* $\delta > 0$.

In words. If the observer can see an account's exact equity, a single test trade settles the question, however close the two candidate markets are. The hard part of the problem is therefore not the market itself but what the observer is allowed to see.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 6).

4.4 Capital paid by the probe trader

Lemma 4.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{15}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (12) achieves error α and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are δ^{-4}.

Lemma 4.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of Appendix D. Its explicit constant is very conservative, because it scales as $m/K_*^2 \propto \epsilon^7 (1-S)^5$: at the calibrated parameters it is 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as a lemma; Appendix D computes the sharp constant, within 5% of a feasible design at every separation.

4.5 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar\eta = 0.2931$ and $\delta = 0.0348$. With equal

weights, the economy is observationally the split pair of Theorem 4.1 around $\bar{\eta}$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q\delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar{\eta}$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: the spectral lower bound of Theorem D.2, evaluated at this pair ($\bar{\eta} = 0.2931$, $\delta = 0.0348$), gives about 77 par units for the optimal flat probe (rescaling the $\bar{\eta} = 0.4$ constant by δ^{-4} alone would give 262, because the constant also depends on $\bar{\eta}$). Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly 1.2×10^5 USDT and the naive pulse about 3×10^7 USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 5.4).

4.6 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (7) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\big(\sqrt{s^2 + 4s} - s\big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and η is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

Proposition 4.6 (Identification in economic primitives)**.** *Consider K Kalman desks (Lemma 2.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$, and write $c_- = \eta'(\bar{s})$ and $c_+ = \eta'(\underline{s})$.*

(i) Prices and aggregate order flow depend on the desks only through the measure

$$\sum_i q_i \, \eta(s_i) \, \delta_{1-\eta(s_i)}.$$

Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.

(ii) Fix a centre s_c and let the ratios lie in $s_c + \Delta s\,[-1,1] \subset [\underline{s}, \bar{s}]$. Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order Δs, or its weights $q_i\eta(s_i)$, from N noisy reset experiments with worst-case error probability below $1/4$ requires $N \geq c\,\sigma^2(\Delta s)^{-(4K-2)}$; on separated configurations, Prony's method of Theorem 5.5 composed with η^{-1} achieves this order. The constants are those of Theorems 5.4 and 5.5 multiplied by powers of c_- and c_+.

In words. What prices reveal about Kalman-filter desks is the mix of how good their information is (their signal-to-noise ratios), weighted by how hard they trade. Expected returns and trading costs matter only through their ratio. Pinning down that mix carries the same steep cost as separating memory rates.

Proof. (i) By Lemma 2.1(i) desk i is a cohort with memory rate $1-\eta(s_i)$ and weight q_i, so Theorem 3.1 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through q_i. (ii) On $[\underline{s}, \bar{s}]$, η is a C^1 bijection onto its image with $c_- \leq \eta' \leq c_+$. Lower bound: take the two economies of Theorem 5.4 with centre $r_c = 1-\eta(s_c)$ and rate spacing $\Delta\eta = c_-\Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1,1]$, so they are two populations of Kalman desks in the class, and the bound $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)}\sigma^2(\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because η is c_+-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates,

which Theorem 5.4 bounds below by a constant times $\Delta\eta = c_-\Delta s$. Upper bound: ratios in $s_c + \Delta s\,[-1, 1]$ with spacing at least $\gamma\Delta s$ have gains in $\eta(s_c) + c_+\Delta s\,[-1, 1]$ with spacing at least $c_-\gamma\Delta s$, a class of the form required by Theorem 5.5 with rescaled separation $\gamma c_-/c_+$; Prony's method recovers the rates and weights at that theorem's rate, and η^{-1}, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

Proposition 4.7 (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio, $q = \varphi(s)$ with φ positive and C^2 on $[\underline{s}, \bar{s}]$ (for a mean–variance desk, $\varphi(s)$ increases with s), and let $n_i \geq 0$ denote the mass of desks with ratio s_i (a divisible population, not an integer desk count). Write $s(\eta)$ for the inverse gain map and $w(\eta) = \varphi(s(\eta))\,\eta$ for the weight per desk as a function of its gain.*

(i) Unknown numbers of desks. *The merged measure is $\sum_i n_i\, w(\eta(s_i))\,\delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1-\eta(\bar{s}), 1-\eta(\underline{s})]$ is of this form, with $n_i = m_i/w(\eta_i)$ for an atom of mass m_i at $1-\eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition 4.6(ii) holds unchanged.*

(ii) Known numbers of desks. *If the n_i are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at $\bar{\eta} \pm \delta$ with n desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2[w(\eta)(1-\eta)^m]_{\bar{\eta}} + o(\delta^2)$, so separating them needs $N \geq c\,\sigma^2\delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

(iii) Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map T from configurations $(s_1, \dots, s_K)$ to perceived ratios that is continuous on a box $\prod_i [\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition 4.6 apply at it.*

Proof. (i) Desk type i is a cohort with gain $\eta_i = \eta(s_i)$, memory rate $1 - \eta_i$ and total weight $n_i \varphi(s_i)$, so it contributes $n_i \varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1 - \eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i / w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1 - \eta)^m$; the tap difference is $n[g_m(\bar{\eta} + \delta) + g_m(\bar{\eta} - \delta) - 2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta}) + o(\delta^2)$, with $\sum_m \sup |g_m''|^2 < \infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over N experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how many desks of each kind there are, and even then the simplest separation problem keeps its δ^{-4} order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map T from a model of the desks' learning.

4.7 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order δ^{-4} blocks (Internet Appendix, Proposition C.2), the same order as the passive Whittle bound of Section 5. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Anonymous, 2026); cohort non-identification alone does not establish that the worst

relevant cost exceeds the informed one.

5 Unknown mixtures and passive data

Theorem 4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

5.1 Why naive pole fitting fails

Because the memory rates are zeros of H (Theorem 3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - ah_t$; (2) form a Hankel matrix of g and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of G; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Appendix C.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Appendix C.3); with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

5.2 Local information: the Cramér–Rao benchmark

Proposition 5.1 (Cramér–Rao benchmark)**.** *Let θ collect the weights $w_1, \dots, w_K$ and rates $\eta_1, \dots, \eta_K$. Suppose the impulse response $h_t(\theta)$, $t \le T$, is observed with i.i.d. Gaussian error from N reset probes of size V_{probe} against innovation noise σ_p, so the error has standard deviation $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. With Jacobian $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, any regular unbiased estimator has $\mathrm{Var}(\hat{w}_k) \ge \sigma_h^2[F_{\text{unit}}^{-1}]_{kk}$, and a 95% interval of half-width* tol *requires*

$$N^* \ge \Big(\frac{1.96}{\text{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\text{probe}}}\Big)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{16}$$

For $K = 2$ with spacing $\Delta\eta$, $[F_{\text{unit}}^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$, so $N^* = \Theta((\Delta\eta)^{-6})$.

In words. This gives the number of test trades any well-behaved estimator needs to pin down one group's trading weight to a stated precision. Each extra group with a nearby memory raises the number sharply, as Table 7 shows.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table 7), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \leq 50$) unbiased estimators cannot resolve close weights.

5.3 A global two-point bound for any estimator

The Cramér–Rao benchmark of Proposition 5.1 applies only to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem 5.2 (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem w_k denotes cohort k's trading intensity q_k, not the merged weight $q_k\eta_k$ of Section 2, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar{\eta} - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{17}$$

(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k\eta_k^m$ for $m \in \{0, 1, 2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{18}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight q, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,

$$\sup_{\theta} P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{19}$$

so failure probability at most 0.05 *requires* $N \geq 4\sigma_h^2 z_{0.95}^2 / \|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. *For* $\Delta\eta \geq 2\tau q/(q + 2\tau)$ *(*≈ 0.018 *at the stated values) the same bound holds without any labelling convention, for the permutation-invariant loss*

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|\big),$$

because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1+\Delta\eta/q),\, (1+2\tau/q)\Delta\eta\} \geq 2\tau$. *For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem 5.4.*

(iii) Conversely, the matching upper bound $N = O((\Delta\eta)^{-6})$ *on rich separated two-cohort classes is Theorem 5.5 with* $K = 2$. *On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i \eta_i r_i^m$, *and perturbation analysis gives the same order.*

In words. For two unknown groups, no method can tell the two candidate markets apart reliably with fewer than a number of experiments of order $(\Delta\eta)^{-6}$, and a practical method (Prony's method, Theorem 5.5) achieves this order. Halving the spacing multiplies the data needed by 64.

We recomputed the two-point quantities exactly (Table 8): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6

resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table 8).

Remark 5.3 (The hard directions carry no risk). The two economies of Theorem 5.2 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1-a-\kappa\sum_i q_i)$ is identical, 2.353 at the calibration of Table 8: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Anonymous, 2026). Identification and safety ask different questions of the same data.

5.4 Many unknown cohorts

Theorem 5.4 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1-r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1-r_c-\Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes $T \geq 2K$ noisy taps ($T = \infty$ allowed); the bound below uses the full response norm, which dominates any finite prefix. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1,1]$, positive weights, and merged measures whose moments of order $0, \ldots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2(\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 5.5 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which Theorem E.1 matches.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K-2$ exactly. For $K=2$ it reproduces the order of Theorem 5.2. Table 9 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K=2,3,4$. The shrinking gap explains why constants deteriorate with K on top of the exponent.

What counts as an experiment. In Theorem 5.4 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and Appendix D prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 6).

Table 10 and Figure 3 confirm the order numerically.

Theorem 5.5 (Matching upper bound for every K). *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j}|s_i - s_j| \geq \gamma > 0$, and weights in $[\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to*

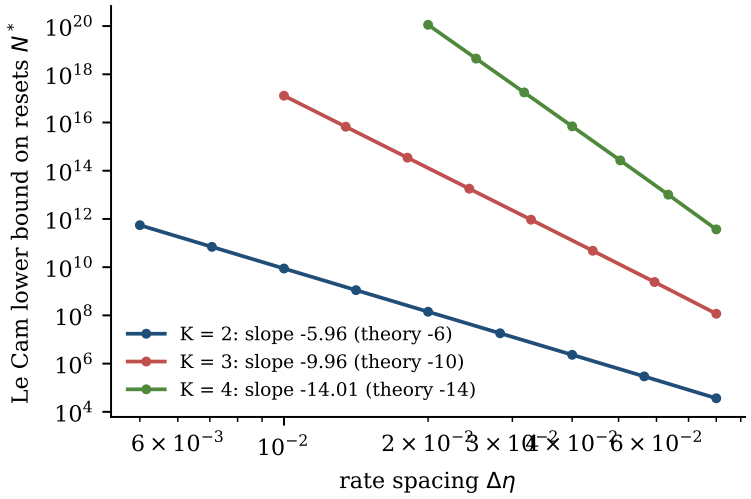


Figure 3: Two-point lower bound on the number of reset experiments needed to separate K-cohort economies, as a function of the rate spacing.

the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with

$$\max_i |\hat{w}_i - w_i| \;\le\; C\sqrt{\log(4K/\beta)}\,\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence, On any such class that contains the Gauss and Gauss–Radau pair of Theorem 5.4 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$. On smaller classes the upper bound need not be tight: if the rates are known, for example, the weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = ah_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \ldots, k_{2K-1}$ from $h_1, \ldots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1, \ldots, h_{m+1}$ with coefficients depending on (a, κ) (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1\le t\le 2K} |\bar{h}_t - h_t| \le \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union

bound over the $2K$ estimated taps give $P(E) \geq 1 - \beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m |\hat{k}_m - k_m| \leq L_h \sigma N^{-1/2} \sqrt{2 \log(4K/\beta)}$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j \leq m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \ldots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \neq y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K - 1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \leq C_1 \sqrt{\log(4K/\beta)}\, \epsilon_N$ with C_1 equal to $\sqrt{2} L_h$ times the norm of the centring map; and because $\sigma N^{-1/2} \sqrt{2 \log(4K/\beta)} \leq \sqrt{2}\, \epsilon_N \sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \leq 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C \sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = L C_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C \sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see

Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 5.4 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

In words. With K unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 5.4 shows that no method does better, and this theorem shows that Prony's method achieves it.

Table 11 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), and Katz et al. (2023) show that Prony's method attains the min-max bound for clustered exponents; here the nodes are real and are first extracted from the closed loop, and it reproduces the ESPRIT slope of -6.15 reported for $K = 2$ in the earlier version.

5.5 Do the rates survive real market noise?

Theorems 5.4 and 5.5 are proved for observation errors that are independent and Gaussian, and Table 11 checks them with Gaussian noise. Real price changes are different: one-minute returns have heavy tails, although they are almost uncorrelated from one minute to the next. To see whether the rates depend on the Gaussian assumption, we ran a pre-specified test (opened once) in which the trader groups are synthetic, with known memory spacing, and only the noise is real. Each reset experiment returns the price response to a unit impulse plus a window of consecutive one-minute log-price changes of Binance BTCUSDT, demeaned and scaled to unit variance. The changes come from 21 days, the 6th of each month from January 2025 to September 2026, that are used nowhere else in this project; their kurtosis

is 19.4, against 3 for Gaussian noise. With class tagging, the experiment also returns each group's flow response, observed through independent windows of the same noise. A control arm uses i.i.d. Gaussian noise of the same variance. For each spacing, N^* is the smallest number of experiments at which every recovered weight lies within 25% of the smallest true weight in at least 75% of 200 replications, using Prony's method on prices alone or the per-group regression of Proposition 6.1 with tagging. The noise level and the spacing grids were fixed on Gaussian noise alone, before any real noise was downloaded.

Both pre-specified hypotheses hold, so the test passes (Table 12). With real noise, the price-only requirement for two groups rises from 64 experiments at $\Delta\eta = 0.5$ to 370,728 at $\Delta\eta = 0.125$, a factor of about 5,800, and the fitted slope is -6.01; for three groups it is -9.81. Over the same grid the tagged requirement stays between 256 and 724 experiments, and it falls slightly rather than rises as the spacing shrinks. At every spacing the real-noise requirement is within one step of the grid (a factor $\sqrt{2}$) of the Gaussian control.

The test has a limit, stated in its specification. Averages of N real windows are resampled exactly up to $N = 4{,}096$; beyond that they are drawn as Gaussian vectors with the real windows' covariance, as the central limit theorem suggests. Real one-minute changes are almost uncorrelated (lag-one autocorrelation between -0.007 and 0.009 within the windows), so beyond 4,096 the real and Gaussian arms differ little by construction. The direct evidence about heavy tails therefore comes from the points in the exact range: every tagged requirement, and the price-only requirements for two groups at the three widest spacings (64, 362 and 1,448 experiments, against 64, 362 and 1,024 with Gaussian noise). The other four price-only points for two groups, and all points for three groups, check the rate with the real noise's variance and covariance but not its tails. Averaging should make heavy tails matter less as N grows, but the test does not show this.

In words. Where real Bitcoin price noise could be used directly, it needs about as many experiments as Gaussian noise, and over the whole grid halving the spacing between two

groups still multiplies the number of experiments by about 64, while class-tagged records keep the requirement flat. At the largest sample sizes the test relies on a normal approximation to the real noise.

5.6 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m \, d\mu(r)$ with $\rho < 1$, where μ is any nonnegative measure of total mass W: a finite or infinite mixture of exponential memories with merged weight W. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a + \kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \int f \, d(\mu - \mu') : \ f \text{ 1-Lipschitz on } [0,\rho], \ f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly W. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree L, and let $c_* = \inf_{L \geq 1} L \, E_L(|x|)$. Bernstein (1914) showed that $L \, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table 13).

Lemma 5.6 (Moment-matched cohort pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0, \rho]$, each with exactly n atoms, whose moments of orders $0, \ldots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho \, E_{2n-2}(|x|) \geq c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Appendix A). Each measure is an n-atom merged measure in the class; one atom may sit at rate 0 (a cohort with $\eta = 1$), which the class allows.

Theorem 5.7 (Minimax rate for general memory)**.** *Let N independent reset experiments observe the impulse response with Gaussian noise of variance σ^2 per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

(i) Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. Whenever $\Lambda_N > 4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu) \geq \frac{c_*\,\rho W \log(1/\rho)}{\Lambda_N}\Big) \;\geq\; \frac{1}{4e},$$

and the bound already holds over n-cohort economies, for a finite n that grows with N.

(ii) Upper bound. *With a and κ known, invert the averaged response into kernel values $\hat{k}_0,\dots,\hat{k}_L$, with $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, and let $\hat{\mu}$ be any measure on $[0,\rho]$ minimizing $\max_{j\leq L}|\int r^j d\hat{\mu} - \hat{k}_j|$ (a linear program). For every $\beta\in(0,1)$ there is N_0 depending only on $(\rho, W, a, \kappa, \beta)$ such that for $N/\sigma^2 \geq N_0$,*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu) > \frac{3\pi\rho W \log B(\rho)}{\log(N/\sigma^2)}\Big) \;\leq\; \beta.$$

Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$, and reaching accuracy w requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$.

Proof sketch. (i) Put the measures of Lemma 5.6, scaled to mass W, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int(1-rz)^{-1}d(\mu_A-\mu_B) = z^{2n-1}\int r^{2n-1}(1-rz)^{-1}d(\mu_A-\mu_B)$; on $|z|=1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A-k^B\|_2$ by the same quantity. The response difference is H_AH_B times the shifted kernel difference, so $\|h^A-h^B\|_2 \leq C_h\|k^A-k^B\|_2$. Choosing the smallest n for which the Kullback–Leibler divergence of the N-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz f with $f(0)=0$, Jackson's theorem gives a polynomial q of degree L with $\sup|f-q| \leq \pi\rho/(4(L+1))$,

and expanding q in Chebyshev polynomials of $2r/\rho - 1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu}, \mu) \le \pi\rho(W + \delta)/(2(L + 1)) + 2\delta\rho B(\rho)^{L+1}$, where δ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1 - az - \kappa/H(z))/(\kappa z)$, whose error is at most $2((1 + s)/\kappa)^2(L + 2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2} L\sqrt{\log L})$. The choice of L makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4} L\sqrt{\log L})$, negligible against the first term. Full proofs are in Appendix A. □

In words. If trader memory is a general mixture of exponentials rather than a few clean ones, accuracy improves only with the logarithm of the data: to halve the error, the data must be squared. The few-groups model of the previous results is therefore the optimistic case.

Theorem 5.7 replaces a polynomial barrier with a logarithmic one. With exponential memory and K known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts is also known (Theorem E.1). The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly (ρ^{2n-1} decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0, \rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table 13 checks the construction and the estimator. The pairs of Lemma 5.6 have exactly n atoms, agree in all moments below order $2n - 1$ to 10^{-61}, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about

0.2 per atom at $\rho = 0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for N from 10^4 to 10^{24}: a 10^{20}-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

5.7 Passive data

Theorem 5.8 (Passive identification: minimum phase)**.** *In a stable economy with $0 < a < 1$, distinct $0 < r_i < 1$ and nonzero merged weights, the zeros of H are the memory rates r_i together with a structural zero at $z = 0$, and H is minimum phase. Degenerate anchor and zero-rate cases require the cancellations described in the merged-equivalence proof. With a single price innovation, the price spectrum identifies H up to scale, hence the memory rates; with κ and a known, as in Theorem 3.1, it identifies the merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 5.9 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem 5.2, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) *For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 5.4,* $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of cohorts. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation; so the orders are the right ones; we do not claim that this attainment is uniform as the separation shrinks.

In words. Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

The proof (Appendix A) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = z^{-1}H_1H_0(G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14.

Passive surveillance is possible in principle but needs at least the same polynomial order of observations as active probing needs experiments. Proposition C.2 is the exact finite-sample version for the known pair.

6 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.
- I_1: prices and aggregated executed flow.
- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition 4.4), also with reports rounded to 0.01.
- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 14 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 6.1 (Tagged flow removes the barrier)**.** *Let the price be driven by i.i.d. innovations of variance σ_ξ^2 in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose cohort i has merged weight $w_i > 0$ and the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. measurement noise of variance σ_f^2 independent of prices. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator*

of $\beta_i = (r_i, w_i)$ *with instruments* $Z_t = (x_{t-1}, x_t)^\top$ *satisfies* $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,

$$V_i = \Sigma_i^{-1}\Omega_i\Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2\big[(1+r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)\big], \quad (20)$$

where $A_m = \sum_{s\geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ *and* $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. *The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \;>\; 0. \quad (21)$$

V_i *depends only on cohort* i*'s own* (w_i, r_i), σ_f^2 *and the price autocovariances, which are bounded and nondegenerate uniformly on a uniformly stable family with merged weights bounded below, whatever the rate spacing; so the merged weights are estimated at rate* $T^{-1/2}$ *with a constant that does not depend on* $|r_i - r_j|$.

In words. If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (21) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods). In simulation (T = 5,000, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

6.1 How much disclosure is worth

Proposition 6.1 says that tagging removes the barrier; the next result says by how much.

Theorem 6.2 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 5.5: rates*

$r_c + \Delta\eta\, s$ with s in a compact set of γ-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 5.4, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let N_p and N_t be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \le C < \infty$, depending on (σ, σ_f) and the class but not on $\Delta\eta$, such that

(i) to recover the weights to fixed accuracy, $c \le N_t \le C$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$;

(ii) to recover the rates to accuracy $c_0\Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$.

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 5.4 and 5.5. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t - s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \ge \omega_{\min}$, the rates lie in a compact subset of $(0, 1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

In words. This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 6.1 and Theorem 6.2 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972). These rates come from day-demeaned regressions, which bias persistence down (Section 6.5), so the factors are illustrative.

6.2 Disclosure in dollars

This subsection is a hypothetical scale illustration of an outside prober's shadow loss, not a public cost–benefit calculation.

Theorem 6.2 states the value of class-tagged disclosure in samples. Theorems D.2 and 5.4 let us translate that value into a flow of round-trip trading losses for an outside participant. Two adjustments make the figures exact:

(i) *Exact K-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem D.2,

$$L^{*}_{\mathrm{rt}}(K, \Delta\eta) \;=\; \frac{4\sigma^{2} z^{2}_{1-\alpha}}{\sup_{\omega}\big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^{2}/g_{0}(\omega)\big)},$$

directly for the K-point Gauss versus Gauss–Radau economies of Theorem 5.4 and Table 9, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss L^*_{rt} is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over, we report the annualised round-trip trading loss $52 \times L^*_{\mathrm{rt}}$ under weekly renewal (matching the CFTC weekly reporting cycle of Section 6.4). For orientation only, we also record official audit-trail figures of two different kinds; we do not divide one by the other, because a prober's trading loss is a private transfer to other traders calibrated to a Bitcoin perpetual, while an audit trail's budget is a resource cost for equities and options. The first is CAT LLC's own operating budget: $248.8 million for 2025 in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to $228.3 million in May 2025 and reduced further later in 2025, and the revised 2026 Financial and Operating Budget of 31 March 2026, with total expenses of $147,044,869, which we use (CAT LLC, 2026). The second is historical, ex ante estimates of industry-wide compliance costs, which are not operating budgets and cover a different scope: the SEC's 2016 CAT NMS Plan Approval Order estimate of $2.4 billion in initial industry build costs and $1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating $4.0 billion initial and $2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

The figures are a shadow cost of learning by trading, the price an outside participant would pay; a supervisor without an audit trail does not probe, and its cost is detection delay, which Proposition 6.3 models. The cost results require a positive economy: with

signed weights the execution form $\frac{1}{2}v^\top T_h v$ can be indefinite and a round trip can earn money, so none of the dollar figures applies there. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning and not a social cost. It is also a lower bound for deterministic probes only: sequential probing can lower the expected loss by a factor of up to 0.49 (Theorem D.3). Only the observation noise is estimated from data; the structural parameters are illustrative, and the constant is sensitive to them. At $\delta = 0.05$ the numerically evaluated lower bound of Theorem D.2 is 358.8, 179.0, 61.1 and 13.9 par units for $a = 0$, 0.25, 0.5 and 0.7, and 3.9, 19.6, 61.1 and 305.2 for $\bar{\eta} = 0.2$, 0.3, 0.4 and 0.6; at $\Delta\eta = 0.20$ instead of 0.10 the unknown-pair figures fall by a factor of about 64. The Gauss–Radau pair is used in the orientation that gives the larger, still valid, lower bound. The known-pair rows span a rate width of $2\delta = \Delta\eta$, whereas the $K = 2$ Gauss–Radau rows on $r_c + \Delta\eta[-1, 1]$ span $\frac{4}{3}\Delta\eta$. Within that interpretation, Table 15 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant Δ_K turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly renewal of price-only probing costs \$0.32 million per market per year at Roll noise and \$7.79 million at one-second noise. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs \$14.74 million (Roll) or \$356.44 million (one-second) per market per year; with $K = 3$ unknown cohorts at $\Delta\eta = 0.20$ the figures rise to \$41.83 million and \$1.01 billion, and at $K = 3$, $\Delta\eta = 0.10$ or $K = 4$, $\Delta\eta = 0.20$ they run to billions. For orientation only, CAT LLC's revised 2026 budget is \$147.0 million; as explained above, the two numbers measure different things. The renewal frequency is an assumption: annual figures scale with it, so monthly renewal divides them by 4.3 and renewal on each of 252 trading days multiplies them by 4.8.

6.3 When disclosure is worth acquiring

When is disclosure worth its cost? The dollar illustration of Section 6.2 prices disclosure; it does not say when a supervisor should buy it. Remark 5.3 shows that a supervisor

who only needs to bound aggregate price risk does not need disclosure at all: the summed response is identified from public prices at the ordinary rate. Attribution matters when supervision is selective: one algorithm must be stopped or sanctioned while the others keep supplying liquidity, and the law requires attributing the conduct to it at a given error level. A supervisor without an audit trail does not run trading probes; it waits for public data to accumulate, and the cost of waiting is the harm done in the meantime.

Consider M markets, a supervisory cycle of $\bar{T}$ periods, and in each market one algorithm whose conduct does harm $\ell > 0$ per period until it is attributed. With prices alone, attribution at the required error level takes $T_p(\Delta\eta) = c_p(\Delta\eta)^{-(4K-2)}$ periods; with class-tagged records it takes $T_t(\Delta\eta) = c_t(\Delta\eta)^{-2}$ periods, the orders of Theorems 5.9 and 6.2 with constants $c_p, c_t > 0$. We take the attribution times to be of exactly these orders, which is a modelling assumption: Theorem 5.9 gives only a lower bound for prices, and if price-only attribution is slower, G below is larger and disclosure is chosen at least on the band the proposition describes. $T_t \propto (\Delta\eta)^{-2}$ is the rate target; for weights at fixed accuracy T_t does not grow as $\Delta\eta$ shrinks, and the lower edge of the band disappears. Attribution that would take longer than the cycle does not happen within it. An audit trail costs F per cycle to operate and c per market to comply with, including any information leakage. The supervisor's expected loss is therefore $\mathcal{L}_{\text{price}} = M\ell \min\{\bar{T}, T_p\}$ without disclosure and $\mathcal{L}_{\text{disc}} = F + Mc + M\ell \min\{\bar{T}, T_t\}$ with it. Write $\Delta\eta_t = (c_t/\bar{T})^{1/2}$ and $\Delta\eta_p = (c_p/\bar{T})^{1/(4K-2)}$ for the spacings at which tagged and price-only attribution just fit within the cycle.

Proposition 6.3 (When an audit trail is chosen)**.** *Let $K \geq 2$ and $\Delta\eta_t < \Delta\eta_p$, and let $G(\Delta\eta) = \min\{\bar{T}, T_p(\Delta\eta)\} - \min\{\bar{T}, T_t(\Delta\eta)\}$ be the attribution delay that disclosure saves.*

(i) The supervisor strictly prefers disclosure if and only if $F/M + c < \ell\, G(\Delta\eta)$.

(ii) G vanishes for $\Delta\eta \leq \Delta\eta_t$, increases on $(\Delta\eta_t, \Delta\eta_p]$ to its maximum $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$, decreases wherever it is positive beyond $\Delta\eta_p$, and is negative for $\Delta\eta > (c_p/c_t)^{1/(4K-4)}$, where it returns toward zero from below. Hence the spacings at which disclosure is chosen form an interval around $\Delta\eta_p$, possibly empty: disclosure is never chosen for

algorithms so similar that even tagged records cannot attribute them within the cycle, nor for algorithms so different that prices attribute them quickly.

(iii) The interval is non-empty if and only if the audit trail covers more than $M^* = F/(\ell G^* - c)$ *markets (with* $M^* = \infty$ *if* $\ell G^* \le c$*), and it widens as* M *grows.*

Part (i) restates the comparison of losses. Part (ii) says that disclosure is valuable in a middle band, where algorithms are close enough that prices cannot attribute them within the cycle but far enough apart that tagged records can; the polynomial gap between the two delays, $(\Delta\eta)^{-(4K-2)}$ against $(\Delta\eta)^{-2}$, is what makes the band wide. Part (iii) is an economies-of-scale result: an audit trail with a fixed operating cost pays only when it spans enough markets. This illustrates a possible economy of scale for market-wide reporting; the cost function is stylised, not estimated from CAT, and the result neither explains CAT's legal architecture nor shows that it is optimal. A supervisor that can trade has a second option, active probing, whose cost is the frontier of Table 15; because that cost grows at the same polynomial order, it shortens price-only attribution only for well-separated algorithms and leaves the comparison for close ones unchanged. The model takes the harm ℓ as given and holds the algorithms' behaviour fixed across regimes; Section 8 returns to both.

A decision in which attribution matters. Proposition 6.3 takes the harm from slow attribution as given. The next result derives such a harm inside the model: a supervisor that must restore stability can act on the whole market, which needs only the combined response, or halt one group, which needs that group's own intensity.

Proposition 6.4 (Market-wide versus targeted intervention)**.** *In the positive, stable economy of Section 2 with known* a *and* κ*, let a supervisor require the closed-loop persistence* $S = a + \kappa \sum_i q_i$ *to be at most* $\bar{s} \in (a, 1)$*. A market-wide measure scales every group's intensity by* $\theta \in (0, 1]$*; a targeted measure halts one group* i*, removing* $q_i = w_i/(1 - r_i)$*.*

(i) The market-wide measure needed, $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$*, depends on the economy*

only through $S = a + \kappa \int (1-r)^{-1} dM(r)$*, which also equals* $1 - \kappa / \sum_j h_j$*; it is estimated at the parametric rate (Remark 5.3).*

(ii) Halting group i *suffices if and only if* $S - \kappa q_i \le \bar{s}$*. Let* A *and* B *be the two* K*-cohort economies of Theorem 5.4 built from the* K*-point Gauss and Gauss–Radau rules, and let* $\omega^G_{\max} \neq \omega^R_{\max}$ *be the largest normalized weights of the two rules. Then* $|S^A - S^B| \le C(\Delta\eta)^{2K-1}$*, while* $|q^A_{\max} - q^B_{\max}| \ge W|\omega^G_{\max} - \omega^R_{\max}|/(1-r_c) - C'\Delta\eta$*. Hence, for small* $\Delta\eta$*, there is an interval of thresholds* $\bar{s}$ *at which halting the largest group restores stability in one economy and not in the other, and any rule based on* N *reset experiments that chooses correctly with probability at least* $3/4$ *in both economies needs, with* $z_{0.75} = \Phi^{-1}(0.75)$*,*

$$N \;\ge\; \Big(\frac{2 z_{0.75}\,\sigma}{\|h^A - h^B\|_2}\Big)^2 \;\ge\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

(iii) With class-tagged flow as in Theorem 6.2, each q_i *is estimated to any fixed accuracy from a number of experiments that does not depend on* $\Delta\eta$*, so for a fixed decision margin* $|S - \kappa q_i - \bar{s}| > 0$ *the targeted decision needs a bounded number of experiments.*

Proof. (i) Scaling all intensities by θ gives persistence $a + \theta(S - a)$, which is at most $\bar{s}$ exactly when $\theta \le (\bar{s} - a)/(S - a)$; $q_i = w_i/(1 - r_i)$ gives the integral form, and $H(1) = \kappa/(1 - S) = \sum_j h_j$ gives the second. (ii) Write $f(r) = (1 - r)^{-1}$ and expand around r_c: $f(r_c + \Delta\eta s) = \sum_m (1 - r_c)^{-(m+1)} (\Delta\eta s)^m$. The two economies have the same total weight W and the same moments of order $0, \dots, 2K - 2$ in $s \in [-1, 1]$, so $|S^A - S^B| \le 2\kappa W \sum_{m \ge 2K-1} (\Delta\eta)^m (1 - r_c)^{-(m+1)} \le C(\Delta\eta)^{2K-1}$ for $\Delta\eta \le (1 - r_c)/2$. Each $q_i = W\omega_i/(1 - r_c - \Delta\eta s_i)$ is within $W\omega_i \Delta\eta/((1 - r_c)(1 - r_c - \Delta\eta))$ of $W\omega_i/(1 - r_c)$, which gives the bound on the largest intensities. Let A be the economy with the larger $q_{\max}$. Then $(S^B - \kappa q^B_{\max}) - (S^A - \kappa q^A_{\max}) > 0$ for small $\Delta\eta$, and any $\bar{s}$ strictly between the two post-halt persistences, and below both S^A and S^B, separates the correct decisions. A rule that is correct with probability at least $3/4$ in both economies is a test between them with both errors at most $1/4$. For Gaussian observations with a common covariance the best such test has error

$\Phi(-\sqrt{N}\|h^A - h^B\|_2/(2\sigma))$, which gives the first inequality, and $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$ (Theorem 5.4) gives the second. (iii) As in the proof of Theorem 6.2, tagged flow gives each (w_i, r_i) with error of order $N^{-1/2}$ uniformly in the spacing, and $\eta_i = 1 - r_i$ is bounded away from zero on the class. □

In words. A supervisor that only needs to cool the whole market can act on public data: the size of a market-wide measure depends on the combined response, which is learned quickly. A supervisor that wants to stop one group and leave the others trading needs that group's own trading intensity, and there are two markets that look almost the same in prices, one in which halting the largest group is enough and one in which it is not. Telling them apart takes the full $(\Delta\eta)^{-(4K-2)}$ cost; class-tagged records reduce it to a bounded number of experiments. Table 16 gives numbers. The largest normalized weights of the two rules differ for every $K \leq 8$ that we checked (for example 0.500 against 0.750 for $K = 2$ and 0.444 against 0.512 for $K = 3$); we do not have a proof that they differ for every K. This proposition is new to this revision and has not been independently reviewed.

What the data can and cannot test. The theorems concern experiments that no outside researcher can run: reset initial states, known price persistence and impact, and either controlled probes or stationary data from a known innovation law. Public data meet none of these exactly. Order flow is endogenous, the fundamental value is unobserved and replaced by returns, observations overlap in time, parameters drift, and the class reports are positions aggregated over many accounts rather than tagged flow. The tests below therefore check the theory's observable prediction, that class records restrict persistence which prices leave unrestricted, under a stated mapping from class positions to cohort states. They do not estimate the theorems' rates, and a pass does not show that class persistence is learning memory.

6.4 Mandated disclosure: the CFTC's trader-class reports

The cleanest test uses disclosure mandated by law. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes a weekly snapshot of Tuesday positions, usually on Friday, aggregated by reporting category: leveraged funds, asset managers/institutional investors and dealers/intermediaries in financial futures, and managed money, producers/merchants/processors/users and swap dealers in commodities. The categories reflect each trader's reported business, not the algorithm that traded. These are published, aggregated class positions, not class-tagged flow. We pre-specified a test before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \,\text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative regresses each week's return on its own twelve lags and fits a two-exponential kernel $\phi_j = b_1 r_1^{j-1} + b_2 r_2^{j-1}$ to the twelve coefficients by least squares over the grid $0 \le r_1 < r_2 \le 0.99$ in steps of 0.02. Weekly futures returns are close to serially uncorrelated, so the fitted weights are close to zero and the rates are weakly identified nuisance parameters (Andrews and Cheng, 2012); that is part of what the wide price-only intervals show. Returns come from Binance spot (Bitcoin, Ether), the S&P 500 index, euro and yen spot rates, minus the change in the 10-year Treasury yield, and front-month crude-oil and natural-gas futures (Yahoo Finance). Only the last two are futures returns; for the other six the basis between spot and the CME contract enters the return regressor; this matters most for the sign of w_c in markets where leveraged funds run basis trades. The comparison is descriptive, not a race between estimators of the same object: the class route fits the persistence of an observed position series, the price route infers latent kernel rates from returns. Intervals come from a 13-week moving-block bootstrap; near a unit root an autoregressive coefficient is estimated at rate T^{-1} rather than $T^{-1/2}$, so the narrow class intervals partly reflect the near-unit-root

persistence of position stocks, such percentile intervals are pointwise and can understate uncertainty, and price-only intervals that fill the admissible range are better read as a sign of weak identification than as calibrated 95% intervals. The pre-specified hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table 17). The class reports pin the fitted persistence rate of each fast class's positions to an interval between 0.013 and 0.106 wide, while the price-only kernel leaves both rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it. A fitted persistence of aggregated positions is not a structural learning memory: it also reflects how slowly positions turn over and how accounts enter and leave a class. In the model with inventory costs (the remark after Lemma 2.1) a position accumulates order flow, so it carries an inventory-adjustment root, close to one when inventory costs are small, on top of the belief root r_i, and a one-lag fit to positions mostly picks up the former. The classes differ as the cohort model allows. Leveraged funds and managed money have the less persistent positions in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and producers; in Treasuries and natural gas the two persistences are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class cannot represent (Barth and Kahn, 2021). A post-hoc check is consistent with it but not decisive: weekly position changes of leveraged funds and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and

Treasuries (-0.37 to -0.67) but not in the euro or yen (-0.04 and -0.05), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal. Negative estimated w_c also lies outside the positive economy of Section 2, so the dollar bounds of Section 6.2 do not apply to these classes; the pre-specified restriction test does not use positivity.

Post hoc robustness. After the registered test we reran the same estimators with three changes, approved before any of them was run and reported without re-scoring the registered result: CME futures prices (BTC, ETH, ES, ZN, 6E and 6J front months) for the six markets that used spot, index or yield prices; the futures version with contract-roll weeks removed by a calendar rule fixed in advance; and 26- and 52-week bootstrap blocks. With futures prices the counts are unchanged (H1 8 of 8, H2 4 of 8, H3 6 of 8), and so is the sign of $\hat{w}_{\text{fast}}$ in every market: the fast class still sells after rises in Bitcoin, Ether and the S&P 500. The spot–futures basis in the registered returns therefore does not produce the negative signs. Longer blocks leave the counts unchanged; Bitcoin's response interval then includes zero. Removing roll weeks leaves H1 and H2 unchanged, lowers H3 to 4 of 8 and turns the Treasury response negative. H3 is lost in Bitcoin and Ether, where the rule removes 46% of the weeks (200 of 436 and 131 of 284): a report is dropped whenever any day of its week touches the five-day expiry window, so with monthly expiries the rule removes about two weeks in four, more than intended, and the fast–slow difference is no longer resolved in the shorter samples. In the equity-index, currency and Treasury markets it removes 15% of the weeks, and in crude oil and natural gas none. The primary result, that class positions pin down a persistence that prices leave open, holds in all seven variants.

Class flows: a pre-specified test that failed. Proposition 6.1 is about tagged flow, while the registered test uses positions. A second pre-specified test (opened once) reads the same reports as flows, the weekly change in the fast class's net position, and estimates $u_t = c + r\,u_{t-1} + w\,\mathrm{ret}_t$ by the instrumental-variables regression of the proposition, with last week's return as the instrument for last week's flow and the same 13-week bootstrap. Its primary hypothesis, that the 95% interval for r is narrower than 0.5 in at least six markets, fails. It holds in three: the S&P 500, crude oil and natural gas, with widths from 0.36 to 0.44. The intervals are 0.57 and 0.60 wide in the euro and the yen, 1.08 in Ether, 3.4 in Bitcoin and 10.8 in Treasuries. The instrument is strong (first-stage $F \geq 10$) in seven markets, all but Treasuries ($F = 4.3$), and the flow response w is nonzero in six, with the same signs as in positions: negative in Ether and the S&P 500, positive in the euro, the yen, crude oil and natural gas. Read as flows, then, the CFTC reports identify how each class's flow responds to prices, but not how persistent that flow is, at the precision the test required. In Proposition 6.1 the precision is of order $T^{-1/2}$ with a constant that grows with the noise in the reported flow (σ_f^2 in (20)). Weekly changes in aggregated positions also move when accounts enter or leave a class, which is noise for this purpose, and two markets have short samples (279 and 428 weeks); we did not test these explanations. The failure limits the claim that published class aggregates implement the tagged-flow contract of Proposition 6.1. It does not bear on the proposition itself, which is a statement about the model.

Together with Table 18, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions restrict persistence rates that prices leave unrestricted; whether that persistence is learning memory or slow turnover of positions, these data do not say.

6.5 Voluntary disclosure: Binance's trader-class ratios

Binance publishes voluntarily, for one venue but every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 6.1, but not homogeneous cohorts: the top-trader position ratio is built from position sizes, the all-account ratio from counts of long and short accounts, the top-trader group is a subset of all accounts, and its membership can change. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 6.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation x_t. For each class c we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio R. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

We begin with the one-hour price kernel of Table 18 and end with the fairer horizon-matched comparison on 30 new markets, which reduces the width ratio to a median of 7.3. Table 18 gives three results. First, tagged class data restrict persistence rates that prices and flow leave essentially unrestricted: with the one-hour price-only kernel, the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing.

The ratios also overstate the gap for a second reason: the price-only kernel spans one hour (twelve five-minute lags), while the tagged half-lives are several hours, so the price-only fit cannot see memories of that length. An exploratory check on the 30 symbols and 105 days of the follow-up test refits the price-only kernel with 48 and 96 lags (four and eight hours) and compares both routes in one unit, the half-life. With 96 lags the price-only slow half-life (median 5.6 hours) reaches the tagged scale (median 4.2 hours). The tagged intervals remain narrower in all 30 symbols, but by a median factor of 7.5 in log half-life rather than 28 to 124, and the price-only interval still reaches a bound of the fit in 25 of 30. The horizon-matched comparison was then pre-specified as a primary hypothesis of a test on 30 symbols never used in this project, before its data were downloaded, and passed: the tagged interval was narrower in 29 of 29 evaluable symbols (median factor 7.3, smallest 2.6), and the price-only interval reached a bound of the fit in 26. A pre-specified secondary of the same test qualifies what the tagged estimate measures. Day-demeaned regressions bias persistence down; corrected by a split-panel jackknife, the median tagged rate is 1.0008 per five minutes, and the corrected interval lies below one in only 7 of 87 class fits. Within a day the class ratios are close to a random walk, so the half-lives of a few hours reported here are largely produced by removing each day's mean. The evidence therefore supports a narrower claim than identification of class memory: positioning data restrict a persistence parameter that prices cannot bound. Whether that persistence reflects learning memory or slow turnover of positions, which theory's tagged contract (class-tagged flow) would distinguish, is not established by these data. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin symbols the classes are often indistinguishable. Third, the pre-specified test fails its 80% criterion, on the part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 5.8 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely.

This is a failure of the prediction as pre-specified, and we report it as one. A post hoc reading, not tested here, is that identification is an asymptotic statement and 42 days of passive thin-market data are far from that regime. Two follow-ups were pre-specified before their data were downloaded, and each ran once. On 105 untouched days in the same 30 markets, price-only intervals were again at least ten times wider than class-tagged ones in all 30 symbols (median ratio 74.9 with the one-hour kernel), but the merged response was significantly different from zero in only 9 symbols, and pinning it down was at least ten times cheaper than separating the memory rates in only 10 of 30 (23 were required): that prediction failed. On 58 further days, the advantage of tagging grew with the sample in 22 of 29 symbols, the number required, but the tagged intervals shrank more slowly than the parametric rate (20 of 29 in the predicted range, 22 required), consistent with drift in class persistence over 21 months. Complete tables for both follow-ups are in the public replication repository.

To test directly whether the memory rates identified by class-tagged positions improve out-of-sample return predictions after large order-flow shocks, we pre-specified a dedicated out-of-sample test before downloading the 22nd-of-month data for January 2025–September 2026 across 19 thin perpetuals (6,137 top-decile five-minute flow shocks on 2024 training days; 4,579 shocks on untouched 2025–2026 test days). On the training sample we estimate each symbol's two memory rates $\hat{r}_1, \hat{r}_2$ either from the class-tagged positions or from the price-only two-exponential fit, construct the filtered memory states $y_{k,t} = \hat{r}_k y_{k,t-1} + (1 - \hat{r}_k)\mathrm{ret}_t$, and forecast the 30-minute post-shock cumulative return $R_{t+1\to t+6}$ out of sample relative to an AR(1) return-and-flow baseline. The primary hypothesis fails: on the untouched test shocks, the price-only memory states yield $\Delta R^2 = -0.199$ percentage points relative to the baseline, the class-tagged memory states yield $\Delta R^2 = -1.288$ points, and the tagged-minus-price-only difference is $\Delta R^2 = -1.089$ points (95% date-block bootstrap CI $[-2.605, +0.376]$, Clark–West $t = 0.96$, $p = 0.1684$). The difference is statistically inconclusive, and there is no evidence of a gain from class data. Likewise, adding the level and one-hour change of the

top-trader and all-account positions to the hourly tail-risk controls of the companion paper leaves out-of-sample R^2 unchanged (-0.19 points, CI $[-0.99, 0.83]$).

After this test failed, we pre-specified a follow-up on untouched 2nd-and-9th-of-month data for January 2025–September 2026 across the same 19 thin perpetuals (12,046 top-decile five-minute flow shocks on 24 training dates in 2024; 9,875 shocks on 42 test dates; test opened once). It asks the narrower question that Theorem 3.1 and Proposition 6.1 answer directly: can prices and flow track the class states themselves? A two-exponential price-and-flow filter ($\alpha_{\text{fast}} = 0.50$, $\alpha_{\text{slow}} = 0.10$ on five-minute returns and signed flow, plus shock return, shock flow and symbol fixed effects) is compared with a filter that adds the current class states and their one-bar changes. Both pre-specified primaries pass by a wide margin (Table 19): adding the class states raises out-of-sample R^2 from -4.8% to 46.0% for the 30-minute divergence between top-trader and all-account positions, and from 4.4% to 54.8% for the top-trader state. As the next paragraph shows, the gain reflects the near-unit-root persistence of position levels, not predictability of future trading.

The gain is almost entirely persistence. In an exploratory decomposition on the same test shocks, an AR(1) on the current level alone accounts for 100.5% and 99.1% of the improvement, and when the target is the 30-minute *change* in positions the class data add -0.26 and $+1.41$ points. Because the class filter observes the series it forecasts, this is a weak test: it confirms that prices and flow cannot reconstruct where the classes are positioned, which Theorem 3.1 implies, but it does not show that class data reveal how their positions will move, and neither pre-specified test finds that class data improve return forecasts.

7 Implications for supervision and market design

The results separate two supervisory tasks that are often run together. Monitoring aggregate fragility, for example to set market-wide limits, needs only the combined price response, which public data identify at the ordinary rate however many groups trade (Remark 5.3,

Proposition 6.4(i)). Attributing feedback to particular groups, to apply targeted limits or enforcement, faces the $(\Delta\eta)^{-(4K-2)}$ barrier whenever groups react over similar horizons (Proposition 6.4(ii)). Class-tagged records remove that barrier group by group (Proposition 6.1), and in the model persistent pseudonymous class labels deliver the full statistical gain without personal data.

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M and the closed-loop impulse response, hence the transfer function, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem 5.2 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Disclosure regimes differ sharply in value (Table 14). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 6.1). In our attribution experiment, correctly tagged flow is far more informative than the periodic statements considered here. Calibrated to the fitted day-demeaned persistences of Binance's published class ratios (an illustrative calibration), separating the top-trader group from all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 6.2). The class reports that the CFTC already requires have the same property in regulated markets (Table 17), so the argument concerns

the granularity and frequency of existing disclosure, not a new kind of reporting.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) is designed to capture specified order and reportable-event information for NMS stocks and listed options across exchanges and broker-dealers, from origination to execution or cancellation, linked to a customer identifier and to the broker-dealer that handled the order; its actual scope is governed by Rule 613, the CAT NMS Plan and exemptions. Since February 2025, under SEC exemptive relief, it no longer requires the names, addresses and years of birth of natural-person customers, who remain linked through an identifier derived from a transformed tax identifier (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three conditional ways. First, for the attribution task modelled here, records that report each class's flow separately remove a barrier that, for an observer who learns by probing, costs round-trip trading loss of order δ^{-4}, and they save samples by a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy or $(\Delta\eta)^{-(4K-4)}$ for rates resolved to a fraction of their spacing (Proposition 6.1, Theorem 6.2). This is a measurement value, not a cost–benefit assessment: CAT's budget pays for enforcement, market reconstruction and surveillance, none of which the model measures, and Section 6.2 sets the hypothetical probing loss beside it only as a scale illustration. For estimating class responses in our model a persistent pseudonymous label suffices; this does not settle what enforcement or data-minimization design requires. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so account-level records need not provide the class-homogeneous flow that Proposition 6.1 assumes; an observer who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the same exponent. MiFID II's order-record and algorithmic-governance rules (RTS 24 for venue order records, European Commission, 2017; RTS 6 for firms' algorithmic systems, European Commission, 2017b) are the closer analogue, although the firm-assigned algorithm identifier tracks a distinct strategy within a firm, not its continuously updated parameters. Third, the

audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not public: regulators and authorized self-regulatory organizations have controlled access, while outside desks and researchers generally do not. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem G.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and confidence sets for the transfer function, both of which are identified from public data. These are cheap where identification is expensive (Remark 5.3); the companion paper shows that the summed response can be bounded at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Anonymous, 2026).

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

8 Conclusion and an open problem

In a linear market with exponential-memory cohorts and known impact parameters, prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a numerically evaluated constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, and at least that order of

passive observations; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights match, populations whose learned weights differ are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier under the stated contract, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$ for resolving rates (and $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy), about 333 for the trader classes Binance already publishes in BTCUSDT. The CFTC's class reports restrict a fitted position persistence that prices leave unrestricted in all eight regulated markets we examined; this persistence is not shown to be learning memory, a pre-specified test of what prices do reveal failed, and class-tagged memory states did not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points, inconclusive). Read as weekly flows, the same CFTC reports identify how each class's flow responds to prices but failed a pre-specified test of how precisely they pin down its persistence. With real Bitcoin price noise in place of Gaussian noise, the predicted rates held in a pre-specified test, at the sample sizes where the real noise could be used directly. At realistic noise the capital barrier for a moderate separation is about 10^5 USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

Two open questions remain. The first is economic: Proposition 6.4 gives one stylised decision in which knowing the individual groups changes the right action, but turning the information costs here into a value of disclosure needs a supervisory loss function estimated from real interventions. The second is for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response and the transfer function, and finite-sample confidence sets for them, are identified from public data even when cohorts are

not. The structural matrix J is not: economies with the same merged measure can have different state dimensions, so J is identified only up to the equivalence class of minimal realizations of the same transfer function. A well-posed next question is: for a stated class of stable closed-loop markets, given an estimated transfer function and a finite-sample confidence set, which intervention policy of least cost guarantees a stated bound on future price displacement without recovering individual cohorts? The companion paper uses exactly this interface to build an identification-free pre-trade control.

Declarations

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Author contributions. Withheld for anonymous review.

Data availability. All data are public. Binance market data come from the Binance public data archive (`https://data.binance.vision`): spot and USD-margined perpetual futures candles at one-second and one-minute frequency, futures order-book depth, and the published trader-class long/short ratios (top traders by position and by account, and all accounts); every file is verified against Binance's published checksums. The CFTC Commitments of Traders reports (Traders in Financial Futures and Disaggregated, futures only, through 13 October 2026) come from the CFTC public reporting API. Weekly prices for the CFTC markets (stock index, currencies, Treasury yield and CME futures) come from Yahoo Finance. Processed data and all result files are in the public repository below.

Code availability. All code, pre-specified test specifications, result files and test-unlock counters are in an anonymized replication package (link withheld for review). Its README

lists the scripts in run order, and Appendix I maps every number in the paper to the file that produces it.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. The research ideas are the authors' own. The authors used AI assistants in carrying them out: Claude (Anthropic) was used to analyse data, read, write and debug analysis code, check derivations and numerical results, and draft and edit text. ChatGPT (OpenAI) and Gemini (Google) were used for planning and to critique drafts; these AI critiques are internal checks, not independent peer review. AI tools are not authors. The authors ran and reviewed all code and results, reviewed all text, and take full responsibility for the content.

References

Alfonsi, A., A. Schied, and A. Slynko (2012). Order book resilience, price manipulation, and the positive portfolio problem. *SIAM Journal on Financial Mathematics* 3. doi:10.1137/110822098.

Alfonsi, A., A. Schied, and F. Klöck (2016). Multivariate transient price impact and matrix-valued positive definite functions. *Mathematics of Operations Research* 41, 914–934.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Amilon, H. (2008). Estimation of an adaptive stock market model with heterogeneous agents. *Journal of Empirical Finance* 15, 342–362.

Andrews, D. W. K., and X. Cheng (2012). Estimation and inference with weak, semi-strong, and strong identification. *Econometrica* 80, 2153–2211.

Banerjee, S., and M. Szydlowski (2026). Trading against algorithms: Price dynamics and risk-sharing in a market with Q-learners. Working paper.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Barth, D., and R. J. Kahn (2021). Hedge funds and the Treasury cash-futures disconnect. Office of Financial Research Working Paper 21-01.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1–57.

Bessembinder, H., W. Maxwell, and K. Venkataraman (2006). Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *Journal of Financial Economics* 82, 251–288.

Biais, B., T. Foucault, and S. Moinas (2015). Equilibrium fast trading. *Journal of Financial Economics* 116, 292–313.

Boehmer, E., G. Saar, and L. Yu (2005). Lifting the veil: An analysis of pre-trade transparency at the NYSE. *Journal of Finance* 60, 783–815.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27, 2267–2306.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Castelli, V., and T. M. Cover (1996). The relative value of labeled and unlabeled samples in pattern

recognition with an unknown mixing parameter. *IEEE Transactions on Information Theory* 42, 2102–2117.

Consolidated Audit Trail, LLC (2026). Consolidated Audit Trail, LLC 2026 Financial and Operating Budget, 31 March 2026 (total expenses $147,044,869). https://www.catnmsplan.com/cat-financial-and-operating-budget.

Cespa, G., and X. Vives (2015). The beauty contest and short-term trading. *Journal of Finance* 70, 2099–2154.

Cochrane, D., and G. H. Orcutt (1949). Application of least squares regression to relationships containing auto-correlated error terms. *Journal of the American Statistical Association* 44, 32–61.

Colliard, J.-E., T. Foucault, and S. Lovo (2022). Algorithmic pricing and liquidity in securities markets. CEPR Discussion Paper 17606.

Crouzet, N., I. Dew-Becker, and C. G. Nathanson (2020). On the effects of restricting short-term investment. *Review of Financial Studies* 33, 1–43.

Dou, W. W., I. Goldstein, and Y. Ji (2025). AI-powered trading, algorithmic collusion, and price efficiency. NBER Working Paper 34054.

Duarte, J., and L. Young (2009). Why is PIN priced? *Journal of Financial Economics* 91, 119–138.

Easley, D., N. M. Kiefer, M. O'Hara, and J. B. Paperman (1996). Liquidity, information, and infrequently traded stocks. *Journal of Finance* 51, 1405–1436.

European Securities and Markets Authority (2016). Guidelines on transaction reporting, order record keeping and clock synchronisation under MiFID II. ESMA/2016/1452.

European Commission (2017). Commission Delegated Regulation (EU) 2017/580 of 24 June 2016 supplementing Regulation (EU) No 600/2014 with regard to regulatory technical standards for the maintenance of relevant data relating to orders in financial instruments (RTS 24), Article 2(1)(b)–(c). *Official Journal of the European Union* L 87, 31.3.2017.

European Commission (2017). Commission Delegated Regulation (EU) 2017/589 of 19 July 2016 supplementing Directive 2014/65/EU with regard to regulatory technical standards specifying the organisational requirements of investment firms engaged in algorithmic trading (RTS 6).

Official Journal of the European Union L 87, 31.3.2017.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257–1272.

Forssell, U., and L. Ljung (1999). Closed-loop identification revisited. *Automatica* 35, 1215–1241.

Franke, R., and F. Westerhoff (2012). Structural stochastic volatility in asset pricing dynamics: Estimation and model contest. *Journal of Economic Dynamics and Control* 36, 1193–1211.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance* 10, 749–759.

Glosten, L. R., and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14, 71–100.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance* 46, 179–207.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66, 1–33.

Hua, Y., and T. K. Sarkar (1990). Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 38, 814–824.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218–2247.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Kukacka, J., and J. Barunik (2017). Estimation of financial agent-based models with simulated maximum likelihood. *Journal of Economic Dynamics and Control* 85, 21–45.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. Springer.

Leach, J. C., and A. N. Madhavan (1993). Price experimentation and security market structure. *Review of Financial Studies* 6, 375–404.

Liu, P., and H. Ammari (2022). Super-resolution of positive near-colliding point sources. arXiv preprint arXiv:2212.00536.

Ljung, L. (1999). *System Identification: Theory for the User*, 2nd ed. Prentice Hall, Upper Saddle River, NJ.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Manole, T., and N. Ho (2022). Refined convergence rates for maximum likelihood estimation under finite mixture models. *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.

Moallemi, C. C., B. Park, and B. Van Roy (2008). Strategic execution in the presence of an uninformed arbitrageur. arXiv preprint arXiv:0801.3001.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821–830.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics* 104, 228–250.

Anonymous (2026). Companion paper on transient amplification and pre-trade guards in closed-loop markets. Working paper, withheld for anonymous review.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Pagano, M., and A. Röell (1996). Transparency and liquidity: A comparison of auction and dealer markets with informed trading. *Journal of Finance* 51, 579–611.

Park, B., and B. Van Roy (2015). Adaptive execution: Exploration and learning of price impact.

Operations Research, doi:10.1287/opre.2015.1415.

Rebafka, T., and F. Roueff (2015). Nonparametric estimation of the mixing density using polynomials. arXiv preprint arXiv:1002.4516.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance* 39, 1127–1139.

Roy, R., and T. Kailath (1989). ESPRIT—estimation of signal parameters via rotational invariance techniques. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 37, 984–995.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

U.S. Securities and Exchange Commission (2012). Consolidated Audit Trail; Final Rule (Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238; proposing release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153).

U.S. Securities and Exchange Commission (2016). Joint Industry Plan; Order Approving the National Market System Plan Governing the Consolidated Audit Trail. Release No. 34-79318, November 15, 2016, 81 FR 84696 at 84801, p. 363.

U.S. Securities and Exchange Commission (2024). Notice of Filing of Proposed Rule Change Relating to the 2025 CAT Budget. Release No. 34-101901, December 12, 2024, p. 4 (revised May 2025).

U.S. Securities and Exchange Commission (2025). Order granting exemptive relief from certain requirements of the National Market System Plan governing the Consolidated Audit Trail. Release No. 34-102386, February 10, 2025.

Shahram, M., and P. Milanfar (2005). On the resolvability of sinusoids with nearby frequencies in the presence of noise. *IEEE Transactions on Signal Processing* 53, 2579–2588.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65, 557–586.

Trebbi, F., M. B. Zhang, and M. Simkovic (2024). The cost of regulatory compliance in the United States. Working paper.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.

Tremacoldi-Rossi, P. (2025). Manipulating algorithmic markets. Working paper.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65, 4753–4768.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333–348.

Wagenmaker, A., and K. Jamieson (2020). Active learning for identification of linear dynamical systems. In *Proceedings of the 33rd Conference on Learning Theory*, PMLR 125, 3487–3582.

Wagenmaker, A. J., M. Simchowitz, and K. Jamieson (2021). Task-optimal exploration in linear dynamical systems. In *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139, 10641–10652.

Wellman, M. P. (2006). Methods for empirical game-theoretic analysis. In *Proceedings of the 21st National Conference on Artificial Intelligence (AAAI-06)*, 1552–1555.

Whittle, P. (1953). Estimation and information in stationary time series. *Arkiv för Matematik* 2, 423–434.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.

A Proofs

Lemma 2.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (7), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2}H(x_t-\theta)^2 = -H(x_t-\theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Lemma 2.3. Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0,1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Theorem 3.1. With zero initial states, $y_{i,t} = \sum_{s\le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s\le t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j\neq i}(z - r_j)$, as in (4); at a retained r_i the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j\neq i}(r_i - r_j) \neq 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The remaining numerator factor z cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\operatorname{supp}(M) \cap (0,1)|$ for $a' \neq 0$ and $|\operatorname{supp}(M) \cap (0,1)|$ for $a' = 0$ (an atom at

rate 0, a cohort with $\eta = 1$, contributes the constant $M(\{0\})$ to G and no pole, so it shifts a to $a' = a + \kappa M(\{0\})$ and the argument above applies with a') (numerically, the rank of the Hankel matrix of $(h_j)_{j\geq 1}$ is $K+1$ and K respectively for $K = 1, 2, 3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z-1+\bar{\eta})\big((z-1+\bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least ϵ. Uniform stability gives $M_H = \sup|H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1}H_\delta H_0(G_\delta - G_0)$, whose factor z^{-1} has modulus one on the circle, gives an ℓ_2 gain at most $C_{\mathcal{F}}\delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal{F}}^2\delta^4 PT$; the equal-noise KL is at most $C_{\mathcal{F}}^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

Theorem 4.1. In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1-S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1-S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_*\delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_*\delta^2\|v\|_1$, and Gaussian KL with Pinsker gives (10). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \leq V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t|\lambda x_t| \leq |\lambda|\kappa V$, attained by a single pulse.

Lemma 4.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i}v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs

gives $\frac{1}{2}v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \le \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem 4.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \le x_t \le \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (14). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition 4.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Lemma 4.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum |v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1 - 2\lambda\cos\omega + \lambda^2) \ge (1-r)/(1+r)$, so $v^\top T_h v \ge m\|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (15). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

Proposition D.1. Parts (i) and (ii) follow from (22) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for δ; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

Theorems D.2 and D.3. For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2}v^\top T_{h_i} v = \frac{1}{2\pi}\int g_i(\omega)|V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi}\int |\Delta\hat{H}|^2|V|^2 \le \sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \ge 2\sigma z_{1-\alpha}$ gives (26). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \ge \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0 \sum_t (\Delta\mu_t)^2/(2\sigma^2) \le \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (28).

Part (iv). Let G_θ be the $T \times T$ lower-triangular Toeplitz section of H_θ/κ, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \le \sup_\omega |H_\theta|/\kappa \le \bar{M}$ because $|H_\theta(e^{i\omega})| \le \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; κ/H_θ is analytic on and outside the unit circle because H_θ is minimum phase (Theorem 5.8), and Lemma 2.3 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1-\lambda_\ell\cos\omega)/(1-2\lambda_\ell\cos\omega+\lambda_\ell^2) \ge \sum_\ell \omega_\ell/(1+\lambda_\ell) \ge \kappa/2$, so $\|G_\theta^{-1}\| \le 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$ (the smallest singular value of G_θ is at least 0.522 in 762 random positive economies and sections). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top\Sigma_1^{-1}\Delta\mu \le \|(H_1 - H_0)v\|_2^2/(2\sigma_{\mathrm{eff}}^2)$. For the covariance part, the eigenvalues λ_k of $A = \Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)$, and $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \le \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\mathrm{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1\Delta G^\top)$ and $\|\Delta G\|_F^2 \le T\|h^\delta - h^0\|_2^2/\kappa^2 \le TK_*^2\delta^4/\kappa^2$ (proof of Theorem 4.1) gives a covariance part of at most $C_w T\delta^4$. The data-processing inequality turns errors (α, β) into $D_{\mathrm{KL}} \ge \mathrm{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (27).

Theorem D.4. Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the G_i-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \ge c_{\mathrm{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \ge 0$; replacing θ by $\theta + \pi$ flips d and s_θ and leaves (29) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$ and

$D = L_i(v) - L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \geq c_{\text{req}} - \sum_{j\leq L-3} \lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j\leq L-3} c_j^2 - L_0(\theta) \;\geq\; \sum_{j\leq L-3} \Big(1 - \frac{\lambda_j}{\lambda(\theta)}\Big) c_j^2 \;\geq\; \Big(1 - \frac{\lambda_{L-3}}{\lambda(\theta)}\Big) \sum_{j\leq L-3} c_j^2 \;\geq 0,$$

and $r^2 \geq L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)} \sum_{j\leq L-3} c_j^2 \geq L_0(\theta)\big(1 - \gamma(\theta)D\big)$. By Hölder's inequality with the sign vector s_θ, $c_{\text{fee}}\|Eu\|_1 \geq c_{\text{fee}} r \|Ed(\theta)\|_1 + c_{\text{fee}} \sum_{j\leq L-3} c_j\, s_\theta^\top E e_j$, and Cauchy–Schwarz with weights $1 - \lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\, W_\perp(\theta)}$. Using $r \geq \sqrt{L_0(1-\gamma D)_+} \geq \sqrt{L_0}\,(1-\gamma D)_+$ and the fact that the fee is nonnegative, with $x = \sqrt{D}$,

$$L_i(v) + c_{\text{fee}}\|v\|_1 \;\geq\; L_0(\theta) + x^2 + \Big[F_{\text{dom}}(\theta)(1-\gamma(\theta)x^2)_+ - \sqrt{W_\perp(\theta)}\, x\Big]_+ \;\geq\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown θ gives (30), and $\Psi \geq 0$, $L_0(\theta) \geq c_{\text{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \leq 1$ with $\gamma F_{\text{dom}} < 1$ the bracket is a convex quadratic in x with minimum $F_{\text{dom}} - W_\perp/(4(1-\gamma F_{\text{dom}}))$, which gives the closed form.

Corollary D.5. Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma_{1,t}^2 - \sigma_{0,t}^2 = \sigma^2 \xi[(h_1 + h_0) * v]_t [\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x - 1 - \log x)$ with $x = \sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

Proposition 5.1. Standard Cramér–Rao for a Gaussian mean model with Jacobian J. For $K = 2$ the columns of J for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of J is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$.

Theorem 5.2. Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (17) gives zero for $m \leq 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with N independent Gaussian experiments gives (19). For the permutation-invariant loss, the identity matching of θ_0 and θ_1 costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom

and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least 2τ, any estimate within τ of one economy in d_∞ is farther than τ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the upper bound is Theorem 5.5; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

Theorem 5.4. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K - 1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the W_1 version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Lemma 5.6. Let $L = 2n - 2$ and let p^* be the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree L, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j\neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial p of degree at most L; multiplying by -1 if necessary, w_i has the sign of $|t_i| - p^*(t_i)$ for every i. Exactly n weights are positive and n negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x + 1)/2$ gives n-atom probability measures π_A (positive part) and π_B (negative part) on $[0, \rho]$ with equal moments of orders $0, \ldots, L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2} \int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because p^* integrates to the same value under both measures, $|x| - p^* = E$ on the support of π_A and $-E$ on that of π_B. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \geq \rho E \geq c_* \rho / L$.

Theorem 5.7. Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa / D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the ℓ_1 norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1 / \kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W \pi_A$ and $\mu_B = W \pi_B$ with π_A, π_B from Lemma 5.6; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1 - rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1} (1 - rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \leq C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the N-sample Gaussian experiments is $N\|h^A - h^B\|_2^2 / (2\sigma^2) \leq (N/\sigma^2)\, 2 C_h^2 W^2 \rho^{2(2n-1)} / (1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \geq \Lambda_N - 2\log(1/\rho)$. Take n the smallest integer with this property and $n \geq 2$; then $2n - 2 < \Lambda_N / (2\log(1/\rho))$ when $\Lambda_N > 4\log(1/\rho)$. Since d is a norm on signed measures and $d(\mu_A, \mu_B) = W\, W_1(\pi_A, \pi_B) \geq W c_* \rho / (2n-2) > 2 c_* \rho W \log(1/\rho) / \Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j \in \{A,B\}} P_j(d(\hat{\mu}, \mu_j) \geq c_* \rho W \log(1/\rho)/\Lambda_N) \geq \frac{1}{4} e^{-1}$.

(ii) *Approximation.* Let f be 1-Lipschitz on $[0, \rho]$ with $f(0) = 0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1, 1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial p of degree L with $\sup|g - p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence $\sup|p| \leq 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \leq 2 \sup|p| \leq 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates f with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \leq \sum_{j \leq L} |b_j|\, \|T_j(2r/\rho - 1)\|_{\text{coef}} \leq 4\rho \sum_{j \leq L} B(\rho)^j \leq \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\text{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\text{coef}} \leq (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of y^i by at most $(1+\alpha)^i$, and $B(\rho) - 1 \geq 4$. On the event $\max_{j \leq L} |\hat{k}_j - k_j| \leq \delta$

the true μ attains objective at most δ, so the minimizer $\hat{\mu}$ satisfies $\max_{j\le L}|\int r^j d\hat{\mu} - k_j| \le 2\delta$ by the triangle inequality. *Moment comparison.* If $\max_{j\le L}|\int r^j d\hat{\mu} - k_j| \le 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\, d(\hat{\mu}-\mu) = \int (f-q)\, d(\hat{\mu}-\mu) + \sum_{j\le L} q_j\Big(\int r^j d\hat{\mu} - k_j\Big) \le \frac{\pi\rho(W+\delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

Kernel recovery. From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \dots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \dots, h_{L+1}$ only. Work in the algebra of power series modulo z^{L+2} with the ℓ_1 norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1\|E\|_1 \le \frac{1}{2}$, then $1/\hat{H} = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \le 2\|1/H\|_1^2\|E\|_1$, so $\max_j |\hat{k}_j - k_j| \le \delta \equiv 2((1+s)/\kappa)^2(L+2)\max_{t\le L+1}|E_t|$. Each E_t is Gaussian with variance σ^2/N, so with probability at least $1-\beta$, $\max_t |E_t| \le \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true μ is feasible for the linear program with value at most δ, so the minimizer satisfies the moment condition above. *Choice of L.* With $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4} L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \le 2\pi\rho W \log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For N/σ^2 larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1\|E\|_1 \le \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W \log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

Theorem 5.9. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi} \big(\varrho - 1 - \log\varrho\big)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1}H_1\Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1}H_1\Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with w_k the cohort weights (any factor of η_k is absorbed in ϕ), where ϕ is

analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding ϕ around $\bar{\eta}$, $\Delta G = \sum_{j \geq 0} \frac{\partial_\eta^j \phi(z,\bar{\eta})}{j!} \Delta \tilde{M}_j$, where $\tilde{M}_j = \sum_k w_k (\eta_k - \bar{\eta})^j$. For the pair, $\Delta \tilde{M}_0 = \Delta \tilde{M}_1 = 0$ and $\Delta \tilde{M}_2 = q\delta^2$; for the moment-matched economies, $\Delta \tilde{M}_j = 0$ for $j \leq 2$ and $\Delta \tilde{M}_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (18); for the K-cohort Gauss and Gauss–Radau economies with rates $\bar{\eta} + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K - 2$ exactly, so $\Delta \tilde{M}_j = W(\Delta\eta)^j \sum_k (\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \leq 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \neq 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta \tilde{M}_j| \leq 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar{\eta}$, uniformly on the circle, gives $|\partial_\eta^j \phi|/j! \leq C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(z^{-1} H_0\, \partial_\eta^{j_0} \phi(\cdot, \bar{\eta})\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $f = z^{-1} H_0 \partial_\eta^j \phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \leq \frac{1}{2}$, $u^2/4 \leq u - \log(1+u) \leq u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-T bound that is uniform in the separation, which we obtain directly. The first T observations are $N(0, \Sigma_j)$ with Σ_j the $T \times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2 |H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem 5.8), so $0 < \underline{f} \leq f_j \leq \bar{f} < \infty$ with constants that do not depend on the separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \geq \underline{f}$ and $\|\Sigma_1 - \Sigma_0\| \leq \sup_\omega |f_1 - f_0| = \sup_\omega f_0 |u| \leq \bar{f} \sup_\omega |u|$. With $A = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2}$, whose eigenvalues lie in $[\underline{f}/\bar{f}, \bar{f}/\underline{f}]$, the inequality $\lambda - 1 - \log\lambda \leq (\lambda - 1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T \| P_0^T) = \tfrac{1}{2} \sum_k (\lambda_k - 1 - \log \lambda_k) \leq \frac{\bar{f}}{2\underline{f}} \|A - I\|_F^2 \leq \frac{\bar{f}}{2\underline{f}}\, T \Big(\frac{\bar{f}}{\underline{f}}\Big)^2 \sup_\omega |u|^2 \leq C\, T \epsilon^2,$$

with C independent of T and of the separation, because $\sup_\omega |u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on T for every T, not only asymptotically. For a fixed pair, $T^{-1} D_{\mathrm{KL}}(P_1^T \| P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

Theorem 5.8. In the delay variable $\lambda = z^{-1}$, $H = \kappa \prod_i (1 - r_i\lambda) / \big[(1 - a\lambda) \prod_i (1 - r_i\lambda) - \kappa\lambda \sum_i w_i \prod_{j \neq i} (1 - r_j\lambda)\big]$, which is (4) divided through by z^{K+1}; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem 5.8. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z \prod_i (z - r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2 |H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 3.1 then gives the merged measure.

Proposition 6.1. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort i, so $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and, by the ergodic theorem for the stationary linear price process, the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i \sum_{s \geq 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1} u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.3), so $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^s$ with $c_\ell = \sigma_\xi^2 \kappa^{-2} \omega_\ell \sum_m \omega_m / (1 - \lambda_\ell \lambda_m) \geq 0$; substituting, $\det \Sigma_i = w_i(\gamma_x(0) A_0 - \gamma_x(1) A_1) = w_i \sum_{\ell,m} c_\ell c_m (1 - \lambda_\ell \lambda_m)/(1 - r_i \lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} = \sum_{t=1}^T (Z_t - r_i Z_{t+1}) \varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance σ_f^2 and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1}) \varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s \in \mathbb{Z}}, \varepsilon_{i,s} : s \leq t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2 [(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (20). On a uniformly stable

family the modes and weights lie in a compact set with $\lambda_\ell \le \bar{\lambda} < 1$, so with $w_i \ge w_{\min} > 0$ we get $\det \Sigma_i \ge w_{\min}(1-\bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem G.1. With J_θ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and P_{11} is affine in s_2 with positive slope, so for each θ a unique $s_2(\theta) \ge 0$ gives $P_{11} = V$ when V exceeds the value at $s_2 = 0$. The one-step price variance is s_1 throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \ge |\theta|$.

Proposition 6.3. (i) $\mathcal{L}_{\text{disc}} < \mathcal{L}_{\text{price}}$ is $F + Mc < M\ell G$. (ii) T_p and T_t are decreasing in $\Delta\eta$, with $T_t \ge \bar{T}$ exactly when $\Delta\eta \le \Delta\eta_t$ and $T_p \ge \bar{T}$ exactly when $\Delta\eta \le \Delta\eta_p$. On $\Delta\eta \le \Delta\eta_t$ both minima equal $\bar{T}$ and $G = 0$. On $(\Delta\eta_t, \Delta\eta_p]$, $G = \bar{T} - c_t(\Delta\eta)^{-2}$, which increases to $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$. On $\Delta\eta > \Delta\eta_p$, $G = c_p(\Delta\eta)^{-(4K-2)} - c_t(\Delta\eta)^{-2} = (\Delta\eta)^{-2}\big[c_p(\Delta\eta)^{-(4K-4)} - c_t\big]$; the bracket is positive exactly when $\Delta\eta < \Delta\eta_0 = (c_p/c_t)^{1/(4K-4)}$, and $G'(\Delta\eta) = (\Delta\eta)^{-3}\big[2c_t - (4K-2)c_p(\Delta\eta)^{-(4K-4)}\big] < 0$ whenever $c_p(\Delta\eta)^{-(4K-4)} > c_t$, because $4K - 2 > 2$. So G decreases wherever it is positive beyond $\Delta\eta_p$, and is non-positive beyond $\Delta\eta_0$. Every superlevel set of G at a positive level is therefore an interval (the negative tail, where G rises back toward zero from below, never reaches a positive level), and the level $(F/M + c)/\ell > 0$ lies above $G = 0$, so the set where (i) holds is an interval contained in $(\Delta\eta_t, \Delta\eta_0)$ that contains $\Delta\eta_p$ whenever it is non-empty. (iii) The interval is non-empty if and only if the level is below the maximum, $F/M + c < \ell G^*$, that is $M > M^*$; the level decreases in M, so the superlevel set grows. □

B Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \le a < 1$

Symbol	Meaning
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i\eta_i$	merged weight
$M = \sum_i w_i\delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa\sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t\rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown cohorts
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j\rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices

C Internet appendix

C.1 A finite-window price-surveillance bound

Theorem C.1 (Finite-window abstention)**.** *Observe* $x_t = sx_{t-1} + w_t$, $w_t \sim N(0,\sigma^2)$, $x_0 = 0$, $t \le T$, *with* $s \in (0,2)$. *Let a possibly randomized rule* Ψ *output Stable, Unstable or Abstain, with* $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ *and* $\sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. *Choose*

$0 < c_0 < 1$ *with* $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. *On* $B_T = (1 - c_0/T, 1 + c_0/T)$,

$$\inf_{s \in B_T} P_s(\Psi = \text{Abstain}) \;\geq\; 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

and for every finite T *the minimax worst-case error of binary rules over* $s \in (0, 2)$ *is at least* $1/2$.

Proof. Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \leq c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \leq \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \leq \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval B_T is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

C.2 Passive distinction of a known pair in finite samples

Proposition C.2 (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations* $w_t \in [-b, b]$ *of variance* $s_w^2 = b^2/3$ *at each price update, observe* $\tilde{x}_t = x_t + \nu_t$ *with* $\nu_t \sim N(0, \sigma^2)$*, use independent three-step reset blocks and no probes, with* $P_* > 3b$. *Let* $b_1 = a + \kappa q \eta$, $b_{2,0} = b_1^2 + \kappa q \eta (1 - \eta)$ *and* $b_{2,\delta} = b_{2,0} - \kappa q \delta^2$. *In each block the covariance of the first and third observations differs by* $\Delta = \kappa q s_w^2 \delta^2$. *With*

$$R = \max_{r \in \{0, \delta\}} \Big\{ (s_w^2 + \sigma^2) \big[s_w^2 (1 + b_1^2 + b_{2,r}^2) + \sigma^2 \big] + s_w^4 b_{2,r}^2 \Big\},$$

the midpoint test of the average product has worst-case error at most α *whenever* $n \geq 4R/(\alpha \Delta^2)$ *blocks are observed: zero controlled quantity and order* δ^{-4} *blocks.*

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for 3.9×10^7 blocks at $\alpha = 0.1$; in simulation the

midpoint test still errs 32–38% of the time with 2×10^5 blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

Proof. In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \le 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \le 0$ for uniform innovations, so it is at most R. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

C.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - a h_t$.
2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.
3. *Transmission zeros:* compute the zeros of the realization, which under known (κ, a) isolate the roots of the denominator of G.
4. *Weights:* with recovered rates, project $u_t = (x_t - a x_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its nonstructural transmission zeros are the memory rates (Theorem 3.1); the known timing zero is discarded, and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 5.4 and 5.5, which is why we do not report it as an estimator of individual cohorts.

C.4 Probing protocols

Table 21 compares four probing protocols at observation horizon $T = 4$ per cycle.

C.5 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 . The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because z contains no inventory. Margin must be computed on a cash-and-inventory ledger.

C.6 Real-noise injection: requirements at each spacing

Table 22 gives the requirements behind Table 12. N^* is read off the grid $2^{j/2}$, so neighbouring values differ by a factor $\sqrt{2}$, and a ratio of $\sqrt{2}$ or $1/\sqrt{2}$ between the two arms is one grid step. Entries above 4,096 use the central-limit draw described in Section 5.5.

C.7 Numerical stress tests of results that are not machine-checked

Four results are proved on paper but not in Lean (Section 5 and Proposition 6.1). We stress-tested each across randomly drawn parameter settings. These are numerical checks, not proofs and not pre-specified tests; no prediction failed (Table 23).

D Internet appendix: Account-level capital

Lemma 4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

Numerical values. Decimal optima and brackets below are numerical evaluations of the stated bounds.

D.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates θ_0 and $\theta_1 = \theta_0 + \delta$ over N periods, submitting probes $V = (V_1, \dots, V_N)^\top$ that generate inventory $I_t = \sum_{s \le t} V_s$. Under contract I_0 (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{22}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi |V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \ge \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \le V_{\mathrm{eff}}$ and $|I_t| \le I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \le B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

Proposition D.1 (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon T per pulse, (δ, B) space splits into three regions.*

(i) Feasible single pulse. *For $\delta \ge \delta_{\min}(T)$ and $B \ge C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

(ii) Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds V_{eff}. Capped pulses each supply divergence $V_{\mathrm{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\mathrm{pulses}}(T, \delta) = \Big\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\mathrm{eff}}^2 \|d_T\|_2^2} \Big\rceil. \tag{23}$$

(iii) Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\mathrm{session}})$*

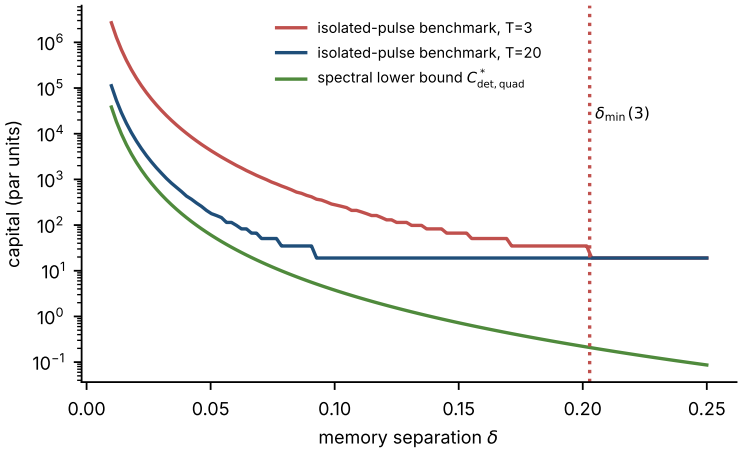


Figure 4: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem D.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

is the root of $V_{\text{eff}}\|d_{T_{\text{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha/\sqrt{\lfloor T_{\text{session}}/T_{\text{cycle}}\rfloor}$*, with small-*$\delta$ *approximation* $\delta_{\min} = \Theta(T_{\text{session}}^{-1/4})$.

In words. A single test trade works only when the groups are far enough apart and the account has enough capital. Closer groups need many capped test trades, and a longer observation window after each trade makes every test trade more informative.

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\text{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\text{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\text{eff}}^2 + \phi P_* V_{\text{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure 4).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

D.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table 24 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.25 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{24}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon L into $L+1$ at identical cost and information, so the minimum is non-increasing in L.

Theorem D.2 (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{25}$$

(i) Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

(ii) Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most α has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\text{det,quad}}(\delta). \tag{26}$$

(iii) Bracket. *At* $\delta = 0.05$, $C^*_{\mathrm{det,quad}} = 61.14$ *at* $\omega^* = 0.2526$*; adding the unavoidable fee* $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta \hat{H}|$ *gives the numerical lower estimate* $C_{\mathrm{det}} \geq 61.41$*. A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

(iv) Process noise. *Let the price also receive the innovations* $w_t \sim N(0, \sigma_w^2)$ *of* (1)*, independent of observation noise* $\nu_t \sim N(0, \sigma_\nu^2)$*, so that from the reset state the observer sees* $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$*,* $t < T$*. For the known pair of Theorem 4.1, put* $\sigma^2_{\mathrm{eff}} = \sigma_\nu^2 + \sigma_w^2/4$*,* $\bar{M} = 1/(1-S)$ *and* $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2 (\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma^4_{\mathrm{eff}})$*. Every deterministic round trip whose test has errors* α *and* β *under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma^2_{\mathrm{eff}} \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta \hat{H}(e^{i\omega})|^2}, \tag{27}$$

with kl *as in Theorem D.3. In-loop noise therefore leaves the* δ^{-4} *capital order intact for every horizon* $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$*; at horizons of order* δ^{-4} *and longer, the noise itself separates the pair without a probe (Theorem 5.9).*

In words. This turns the order into a dollar floor. No round trip that tells the two candidate markets apart at 5% error can lose less than this amount, and a concrete probe comes within 5% of it.

Positivity of g_i follows from Lemma 2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\mathrm{det,quad}}$ across separations (Table 25): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\mathrm{det,quad}}$ between 3.67×10^{-4} and 3.84×10^{-4}, confirming the δ^{-4} law. The fee term, the feasible upper bounds and the adaptive bound of Theorem D.3 are reported in the same table.

The infimum over ω is evaluated on a fine frequency grid; a local refinement agrees to four digits (61.1377 at $\omega^* = 0.2526$), and the reported constant is the larger of the two hypotheses' bounds. "Certified" here means a numerical evaluation of the stated bound, not an interval-arithmetic enclosure.

Theorem D.3 (Adaptive and sequential probing)**.** *Let* $(v_t)_{t \leq \tau}$ *be any non-anticipating policy*

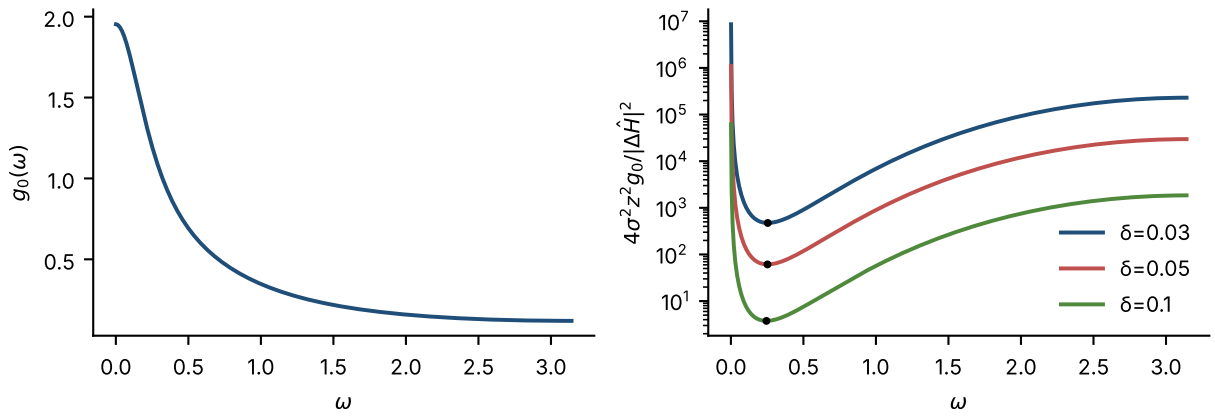


Figure 5: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (26); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

adapted to the price filtration, with an almost surely finite stopping time and $\sum_{t\le\tau} v_t = 0$, *followed by a terminal test* $\psi_\tau \in \{0,1\}$ *with* $P_0(\psi_\tau = 1) \le \alpha$ *and* $P_1(\psi_\tau = 0) \le \beta$. *Then*

$$\mathbb{E}_0[L_0(v)] \;\ge\; 2\sigma^2\,\mathrm{kl}(\alpha, 1-\beta)\inf_{\omega}\frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{28}$$

where $\mathrm{kl}(\alpha,1-\beta) = \alpha\log\frac{\alpha}{1-\beta} + (1-\alpha)\log\frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free loss by at most the factor* $\mathrm{kl}(\alpha,1-\beta)/(2z_{1-\alpha}^2) \approx 0.4897$ *at* $\alpha = \beta = 0.05$, *and cannot alter the* δ^{-4} *order.*

In words. Letting the observer adapt its test trades as data arrive, and stop as soon as it is confident, helps only a little: at the calibration it can cut the expected loss by at most about half. The steep cost is a property of the market, not of a clumsy probing plan.

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a numerical ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under H_0, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before

the test ends. Expected-loss savings are not solvency savings.

Theorem D.4 (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of $L \geq 4$ trading periods, let $E \in \mathbb{R}^{L\times(L-1)}$ be the first-difference basis of flat probes ($\sum_t v_t = 0$), $Q_i = \frac{1}{2}T_{h_i}$ (so that $L_i(v) = v^\top Q_i v$), $A_{\text{diff}} = (H_1 - H_0)^\top(H_1 - H_0)$, and let $G_i = E^\top Q_i E \succ 0$ and $M = E^\top A_{\text{diff}} E \succeq 0$ be the reduced execution-loss and information matrices, with G_i-orthonormal generalized eigenvectors e_j and eigenvalues $0 \leq \lambda_1 \leq \cdots \leq \lambda_{L-3} < \lambda_{L-2} \leq \lambda_{L-1}$; put $c_{\text{req}} = 4\sigma^2 z_{1-\alpha}^2$ and $c_{\text{fee}} = \phi P_*$. For $\theta \in [0,\pi)$ let $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$, $\lambda(\theta) = \lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$, $s_\theta = \operatorname{sign}(Ed(\theta))$, and*

$$F_{\text{dom}}(\theta) = c_{\text{fee}}\sqrt{\frac{c_{\text{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c_{\text{fee}}^2 \sum_{j\leq L-3} \frac{(s_\theta^\top E e_j)^2}{1-\lambda_j/\lambda(\theta)},$$
$$\gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta)-\lambda_{L-3})\,c_{\text{req}}}. \tag{29}$$

Then every flat probe meeting the separation constraint of (24) satisfies

$$L_i(v) + c_{\text{fee}}\|v\|_1 \;\geq\; \min_\theta\Big\{\frac{c_{\text{req}}}{\lambda(\theta)} + \Psi(\theta)\Big\} \;\equiv\; C_{\text{RH}}(L) \;\geq\; \frac{c_{\text{req}}}{\lambda_{L-1}}, \tag{30}$$

$$\Psi(\theta) = \min_{x\geq 0}\Big\{x^2 + \Big[F_{\text{dom}}(\theta)\big(1-\gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\,x\Big]_+\Big\} \geq 0,$$

a piecewise-quadratic minimization in closed form; when $\gamma F_{\text{dom}} < 1$ and the minimizer has $\gamma x^2 \leq 1$, $\Psi \geq \max\big(0, F_{\text{dom}} - W_\perp/(4(1-\gamma F_{\text{dom}}))\big)$, which is what the bound uses.

In words. This is the exact cheapest probe over a finite number of trading periods, fees included. It confirms that the dollar floor of Theorem D.2 is not an artefact of assuming an infinitely long probe.

The bound is computable for any horizon. Table 26 reports its quadratic part $c_{\text{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\text{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective (Table 26 reports

65.02 for ledger capital, which also includes margin), so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or L_1-localized probes are needed at long horizons.

D.3 Robustness: heteroskedastic impact and probing protocols

Corollary D.5 (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma_{i,t}^2(v) = \sigma^2(1 + \xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\text{eff}} = \kappa(1 + \xi_0|v_t|)$*. For capacity-bounded probes,* $D_{\text{KL}}(P_0^v \| P_1^v) \le \frac{1+O(\xi^2 V_{\text{eff}}^2)}{2\sigma^2} \|(H_1 - H_0)v\|_2^2 = O(\|v\|_2^2 \delta^4)$*, and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$*.*

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near -4 in $\log \delta$: -3.90, -3.84, -3.47 and -3.81 over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table 21). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

D.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2 \delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\text{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about 9.9×10^4 USDT, which

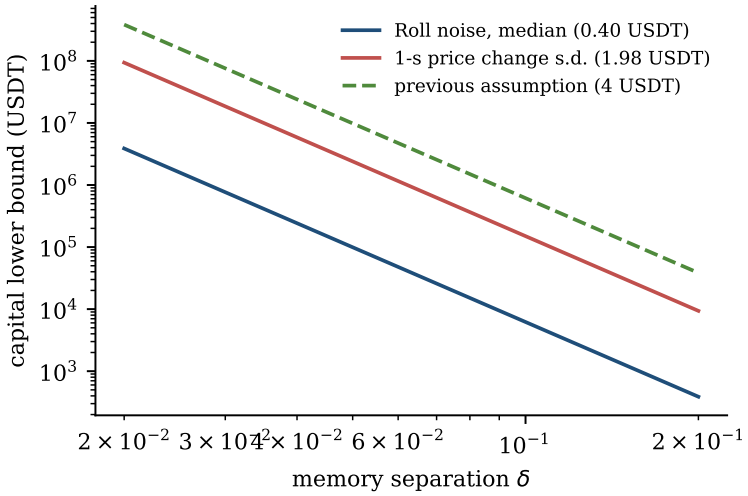


Figure 6: Capital lower bound against memory separation for three observation-noise calibrations.

is not prohibitive for a well-capitalized desk (Table 27). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 5.4, when several cohorts are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

E Internet appendix: Near-colliding rates: a uniform rate for the merged measure

Theorem 5.5 needs the rates to stay γ-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights w and $W - w$ at rates r and $r + \Delta$ and two of weights $W - w$ and w at the same rates have merged measures that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

Theorem E.1 (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most K atoms in $[0, \bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From N reset experiments as in Theorem 5.5, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is C depending only on $(K, \bar{r}, a, \kappa, B)$ such that, with probability at least $1 - \beta$,*

$$W_1(\hat{\bar{M}}, \bar{M}) \;\leq\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \text{uniformly over } \mathcal{M}_K.$$

Hence $N = O(\sigma^2 m_{\min}^{-2}\epsilon^{-(4K-2)})$ experiments give accuracy ϵ, and with the Wasserstein statement of Theorem 5.4 the minimax rate for the merged memory distribution is $N^{-1/(4K-2)}$, uniformly over near-colliding economies.

In words. Even with no lower limit on how close the groups' memories are or how small a group is, the combined memory distribution can still be estimated, and no faster rate is possible. What is lost when groups nearly coincide is the ability to tell them apart, not the ability to describe the combined picture.

Proof. As in the proof of Theorem 5.5, the kernel values $k_0, \ldots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \leq e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$ (the Lipschitz constant is for a bounded neighbourhood of the class; when the noise is too large for that, the right-hand side of the theorem exceeds $\bar{r} \geq W_1$ once C is enlarged, so the bound holds trivially). The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \leq 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \leq k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy $|m_t(\hat{M}) - m_t(M)| \leq 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \leq 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries both normalized measures to distributions on $[-1,1]$ with at most K atoms, multiplies W_1

by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1+2/\bar{r})^{2K-1}$. The moment comparison inequality for K-atomic distributions on $[-1,1]$ (Wu and Yang, 2020), $W_1 \le c_K \delta^{1/(2K-1)}$ when the first $2K-1$ moments differ by at most δ, then gives the bound with $C = c_K(\bar{r}/2)(1+2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem 5.5 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes arbitrarily expensive as rates approach each other (Theorem 5.4) and impossible when they coincide (Theorem 3.1). A supervisor who needs the first can have it at the rate of Theorem 5.4; one who needs the second has to be told the number of cohorts and a separation, or has to obtain tagged data (Section 6).

F Internet appendix: Robustness to nonlinear and adaptive trading rules

The theorems of Sections 3–5 use the linear exponential-memory recursion (2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size A moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1-\eta_i$, whose

gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size η_i; and (5) a *tabular value rule* that updates a 21-bin table at rate η_i, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate η_i, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight q and position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B (δ^{-4} law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as δ^2 and the capital lower bound of Theorem D.2 as δ^{-4}. The saturation threshold A_{sat} is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-specified a second experiment in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate η_i: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and δ^{-4} predictions, not whether learning produces exponential memory. The specification fixed four

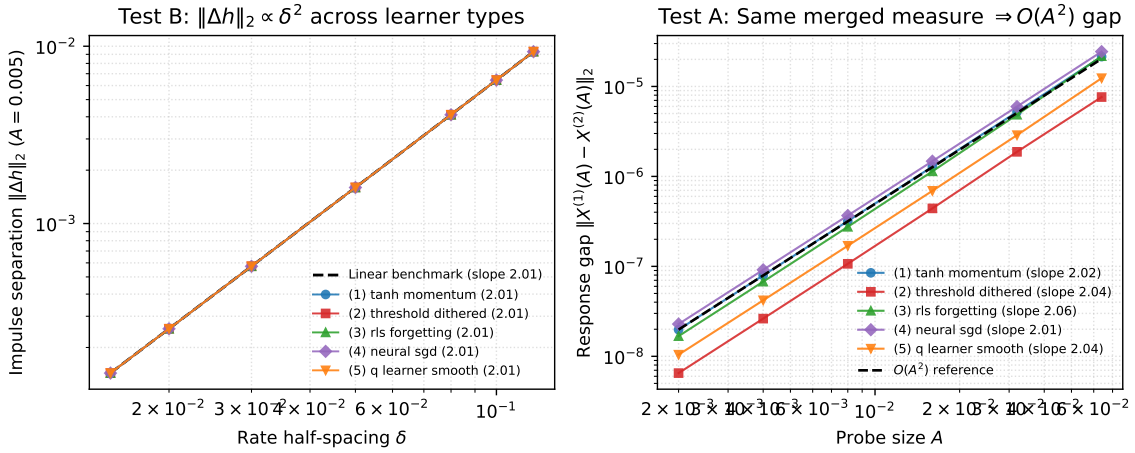


Figure 7: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing δ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size A, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative L_2 response at $A = 0.005$; (c) the capital lower bound to separate trained populations whose memory rates differ by 2δ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the δ^{-4} law fail when learning continues during the probe.

Table 29 and Figure 7 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and capital lower bound scale as δ^2 (slope 2.007) and δ^{-4} (slope -4.001), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most 10^{-5} in relative response at $A = 0.005$, and capital lower bound scales with slopes -4.014 to -3.973,

inside the registered band. The level of capital differs across learners because training sets the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are -1.75, -0.97, -1.62 and -0.38. Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The δ^{-4} law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes -4.02 to -3.98) and the merged-measure gap rises only to 1.1×10^{-3}, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights, not their rates.

G Internet appendix: A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem G.1 (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance V large enough, there is a family of two-state*

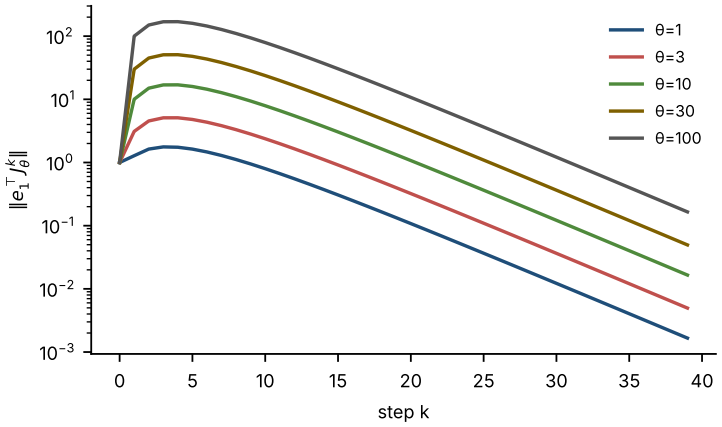


Figure 8: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ *with shock covariance* $\mathrm{diag}(s_1, s_2(\theta))$ *such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain* $\sup_k \|e_1^\top J_\theta^k\|$ *grows without bound in* θ.

In words. Two markets can look identical by the usual risk measures, the same eigenvalues and the same price variance, and yet one of them can produce far larger losses in the short run. Monitoring eigenvalues or volatility alone therefore cannot certify that a market is safe.

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure 8).

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix C.5). Second, in a network of M identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the

Kreiss constant stays at one, as it must for a stable symmetric operator. A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial impact, which also depends on the input and output maps. The positive-cohort no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

H Internet appendix: Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 6.5 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

H.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces p_t^* by a trailing filter. Let p_b be a log price proxy on uniform bars of length Δ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{31}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{32}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let f_t be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{33}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{34}$$

Equation (32) implies that $\hat{J}_{11}$ must be compared with $1-\alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

H.2 Data

H.3 Clock alignment dominates reduced-form feedback

We estimate (33) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table 31), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo

reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

H.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state x_t is the log price deviation from its EMA (span 30 one-second bins), f_t is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table 33 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table 31; the placebo shows only that the filter does not produce it.

H.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta\Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock z_t. Lagged returns are correlated with u_{t-1}, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move y through prices but are independent of u ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table 34).

OLS is badly biased. Cochrane–Orcutt with the true ρ, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated ρ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at

our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from 3.15×10^{-4} to 3.21×10^{-4} (post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table 32 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table 32, column $\hat{Q}_{30}$), consistent with persistent order splitting.

H.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

H.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table 35).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have op-

posite signs, neither is significant, and the pooled effect is -0.00016 with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

I Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source	Kind
Complete monotonicity, 892 economies	`L26_complete_monotonicity`	recomputed
$d_2 = -\kappa^2 q \delta^2$	`P1.7-T1_third_tap`	recomputed
Ledger identity; T_h positive definite	`R2-T1`, `R2-T3`	recomputed
Figure 2	`make_figs2.py`	recomputed
$\delta_{\min}(3)$, Table 25 recomputed columns, 61.14, 29.94	`recompute_theory.py` → `R_P1_theory_constants.json`	recomputed
Pulse benchmarks (271, 58, 12, 11 pulses), Figures 1–4	`make_figs3.py` → `R_P1_pulse_benchmarks.json`	recomputed
Table 6, Table 9, worked example, passive simulation	`recompute_extra.py` → `R_P1_extra.json`	recomputed

Result	Source	Kind
Daily noise (Table 28)	`P1.3-T1_sigma_obs`	real data
Table 8 recomputed columns	same	recomputed
K-cohort slopes (Table 10)	`T1_K_cohort_rate`	recomputed
ESPRIT slope -6.15	`P1.2-T1_esprit_rate`	recomputed
Minimum phase, 1,733 economies; rank 6 of 6	`T2a`, `T2b`	recomputed
Passive slopes -4.21, -5.93	`T3_passive_rate`	recomputed
Tagged-flow slope 0.011	`T4_disclosure`	recomputed
Spectral blindness gains	`T5_spectral_blindness`	recomputed
Noise and capital grid (Table 27)	`P1.3-T1_sigma_obs`	real data
Loop gain and placebo (Table 33)	`P1.6-T1_ema_placebo`	real data
Estimator table (Table 34)	`P1.5-T1`, `T2`, `T3`	recomputed
Data integrity (111 files)	`R2-T9_data_integrity`	audit
Tables 24, 25 (certified, feasible, adaptive columns)	`recompute_p1_capital.py` → `R_P1_capital_ledger.json`	recomputed
Table 14; resolution floors; Table 21	`recompute_p1_protocols.py` → `R_P1_protocols.json`	recomputed
SPRT capital figures	`recompute_p1_sprt.py` → `R_P1_sprt.json`	recomputed
Table 7 (N^*, σ_{crit})	`recompute_crb.py` → `R_P1_crb.json`	recomputed
Table 7 (ML columns), Table 8 (estimators)	`recompute_p1_estimators.py`	recomputed
Symmetric multi-asset network (Appendix G)	`recompute_oct20.py`	recomputed

Result	Source	Kind
Proof audit: K_* bound and d_2 identity (Theorem 4.1), moment gaps (Theorem 5.2), spectral-blindness example (Theorem G.1)	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed
Theorem 5.5, Table 11	`kcohort_upper.py` → `R_P1_kcohort_upper.json`	recomputed
Lem. 5.6, Thm 5.7, Tab. 13	`general_minimax.py`	recomputed
Theorem 6.2 numbers	`disclosure_value.py` → `R_P1_disclosure_value.json`	recomputed
Tables 31, 32, 35	Part 8: `V2_spot_panel`, `V3_fee_events`	real data
Table 18, top panel	Part 4: `X9_disclosure_metrics`	real data
Table 18, bottom panel; class-position forecasts	Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36...)	real data
Table 17	Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35...)	real data
Basis-trade signature; real-data Cochrane–Orcutt	Part 13: `R6`, `R7`	real data (post hoc)
Feedback across 30 perpetuals	Part 4: `X3_T6_cross_section`	real data
Theorem D.4 full bound (Table 26), direct minimization	`hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json`	recomputed

Result	Source	Kind
Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant	`hardening/p1_checks.py` → `R_P1_hardening_checks.json`	recomputed
Stylized nonlinear and trained online rules (Table 29, Figure 7)	`recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…)	recomputed
Disclosure in dollars (Table 15)	`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`	recomputed
Out-of-sample post-shock return test (Section 6.5)	`recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…)	real data
Out-of-sample cohort-state trajectory test (Table 19)	`recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…)	real data
Proposition 6.4, Table 16	`theory/targeted_intervention.py` → `R_P1_targeted_intervention.json`	recomputed

Result	Source	Kind
Table 8, column N^*_{CRB}; Table 7, spacing-0.10 three-cohort row	`breakthrough_experiments/results_exp17_minimax_lecam_bou` `results_exp15_cohort_information.json`	recomputed (earlier run)
Table 26, best feasible probes and $L = 320, 480$	`R_P1_capital_ledger.json`, `R_P1_cquadL.json` (reproduced by `hardening/rh_quad.py`)	recomputed
Table 24, peak margins of the eigenvector probes	`breakthrough_experiments/results_exp14_capital_frontier_`	recomputed (earlier run)
Table 13	`theory/R_P1_general_minimax.json`	recomputed
Table 8 and Table 7, estimator columns	`R_P1_estimators.json`, `R_P1_estimators_crb.json`	recomputed
Theorem 3.1, numerical check of the equivalence	`R2-T5_merged_invariance`	recomputed
Section 6.5 and Table 1: horizon-matched test on 30 new markets (29 of 29, median 7.3, smallest 2.6) and jackknife-corrected persistence (1.0008)	`thesis/evidence/C2_results.json` (specification `thesis/checks/prereg/C2_margin_vs_identify.json`)	real data
Section 6.5: exploratory 48- and 96-lag refit (median 7.5)	`thesis/evidence/pilot3_horizon_matched_results.json`	real data (exploratory)
Section 6.5 and Table 1: 105-day follow-up (median ratio 74.9; 9 identified; 10 of 30, 23 required)	`thesis/evidence/E1v2_results.json`	real data

Result	Source	Kind
Section 6.5 and Table 1: 58-day follow-up (22 of 29; 20 of 29, 22 required)	`thesis/evidence/D1_results.json`	real data
Section 5.5, Tables 12, 22 and 1: real-noise injection (slopes −6.01, −9.81; tagged 256–724)	`p1_injection/injection_test.py`, `p1_injection/results/I1_injection_test.json` (specification `p1_injection/prereg/P1_injection.json`, 596a700d…)	semi-synthetic
Supplement C.7 and Table 23: numerical stress tests	`p1_stress/stress_theorems.py`, `p1_stress/results/S1_stress_tests.json`	numerical check
Theorem 3.1(i), Theorem 5.4 and Proposition 6.1 (machine-checked parts)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedModel.lean`, `Paper1Identification.lean`, `Paper1MomentGap.lean`, `Paper1PoleCount.lean`, `Paper1Tagged.lean`, `Paper1Testing.lean`, `Paper1LowerBound.lean`, `Paper1MatchedPair.lean`	Lean 4 proof
Section 6.4: post hoc robustness (futures prices, roll weeks, 26- and 52-week blocks)	`p11/robustness_cot.py`, `p11res/results/C4R_cot_robustness.json`	real data (post hoc)

Result	Source	Kind
Section 6.4 and Table 1: class-flow test (3 of 8; $F \geq 10$ in 7; $w \neq 0$ in 6)	`p11/tagged_flow_test.py`, `p11res/results/C5F_cot_tagged_flow.json` (specification `p11/prereg_flow/P11F_cot_tagged_flow.json`, 052a63f2…)	real data

Table 5: Map of results

Question	Answer	Where
What do prices identify?	The merged rate–weight measure only	Thm 3.1
Quantity to separate a known pair	$\Theta(\delta^{-2})$, exact error formula	Thm 4.1
Capital to separate a known pair	$\Theta(\delta^{-4})$, counterparty or own funds	Thm 4.3, Lem. 4.5
Constant at the calibration	$[61.41, 64.48]$ par units; $\approx 10^5$ USDT	Thms D.2–D.4, Table 27
Does adaptivity help?	At most by a factor ≈ 0.49 in expected loss	Thm D.3
Unknown mixtures	$(\Delta\eta)^{-(4K-2)}$, tight on rich classes for every K	Thms 5.4, 5.5
General memory	Accuracy $\asymp 1/\log N$, lower and upper bounds	Lem. 5.6, Thm 5.7
Passive data	Identifies M; at least the same polynomial cost for every K	Thms 5.8, 5.9, Prop. C.2
Nonlinear and adaptive rules?	Settled online learners obey δ^{-4}; fast adaptation or thresholds break it	Table 29
What removes the barrier?	Tagged flow, audited equity	Props 4.4, 6.1
Does attribution change a decision?	Yes in a stylised case: a market-wide limit needs only S; whether a targeted halt suffices needs each q_i	Prop. 6.4
What is tagging worth?	Factor $(\Delta\eta)^{-(4K-2)}$ (fixed accuracy) or $(\Delta\eta)^{-(4K-4)}$ (rates); $\approx 333\times$ for BTCUSDT (illustrative)	Thm 6.2
Hypothetical probing loss?	At least \$15M–\$356M a year per market to re-learn two unknown groups weekly at $\Delta\eta = 0.10$; illustrative, not a social cost	Table 15
Do class data restrict class persistence?	Yes: CFTC 8 of 8; Binance 30 of 30 and 29 of 29 new markets (registered); whether it is memory is open	Tables 17, 18
Do the rates survive real price noise?	Yes where it was used directly; large samples use a normal approximation	Table 12
Can eigenvalues certify safety?	No	Thm G.1
What do feedback indices measure?	Mostly filtering and timing; small real feedback	Tables 31, 33
Does feedback track activity?	No: fees move volume 2.6–4.9×, not $\hat{Q}$	Table 35

Table 6: Exact known-pair error: formula versus simulation

Pulse V	Horizon T	$\Phi(-V\|d_T\|_2/2\sigma)$	Simulated error, H_0	Simulated error, H_δ
20	3	0.3745	0.3734	0.3762
40	3	0.2611	0.2599	0.2603
40	10	0.0029	0.0030	0.0027
80	20	1.8×10^{-10}	0	0

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Table 7: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

Scenario	K	N^* probes	σ_{crit}	ML success ($f = 1$)	RMSE/CRB
$\eta = 0.30$	1	0.1	6.05×10^{-3}	94%	1.02
$\Delta\eta = 0.30$	2	47.8	2.31×10^{-4}	94%	1.01
$\Delta\eta = 0.20$	2	597	6.55×10^{-5}	96%	0.98
$\Delta\eta = 0.10$ ($\delta = 0.05$ pair)	2	40,374	7.96×10^{-6}	93%	1.03
$\Delta\eta = 0.05$	2	2,620,320	9.88×10^{-7}	95%	1.03
spacing 0.10	3	17,349,791	3.84×10^{-7}	98%	1.22
spacing 0.20	3	27,038	9.73×10^{-6}	98%	0.88

Tolerance ± 0.01 at 95%. σ_{crit} is the single-probe noise at which one probe suffices; operational single-probe noise is 1.6×10^{-3}. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at σ_{crit}; success means every weight within ± 0.01. Fitted log–log slope of N^* in $\Delta\eta$: -6.04.

Table 8: Two-point sample complexity for two-cohort weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N^*_{CRB}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$, $N = 50$, 1,000 trials per hypothesis: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Estimators: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

Table 9: Gauss and Gauss–Radau rules on $[-1, 1]$ used in Theorem 5.4

K	Gauss nodes / Gauss–Radau nodes	Max gap, orders $< 2K - 1$	Gap at order $2K - 1$
2	± 0.5774 / -1, 0.3333	6×10^{-17}	0.2222
3	0, ± 0.7746 / -1, -0.2899, 0.6899	2×10^{-16}	0.0533
4	± 0.3400, ± 0.8611 / -1, -0.5753, 0.1811, 0.8228	4×10^{-16}	0.0131

All Gauss–Radau weights are positive.

Table 10: Numerical check of Theorem 5.4

	$K = 2$	$K = 3$	$K = 4$
Theoretical exponent of N^*	-6	-10	-14
Fitted log–log slope (small $\Delta\eta$)	-5.96	-9.96	-14.01

Le Cam bound $N^* \propto \sigma^2 / \|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

Table 11: Sample requirement of Prony's method against the lower-bound order

	$K=2$	$K=3$	$K=4$	$K=5$
Lower-bound order $-(4K-2)$	-6	-10	-14	-18
Fitted slope of N^* in $\Delta\eta$ (Prony)	-5.94	-10.00	-13.93	-18.34

Rates $0.7+\Delta\eta\, s$ with s the K-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; N^* is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point. At $K=5$ the fit is numerically sensitive: rerunning the same seeded script with a different numerical library gives slopes -5.94, -10.00, -13.92 and -17.89.

Table 12: Sample requirement with real Bitcoin price noise: fitted slope of $\log N^*$ in $\log \Delta\eta$

	Theory	Real noise	Gaussian control	Pre-specified criterion	Result
Price only, $K=2$ (H1)	-6	-6.01	-6.22	slope in $[-7.5, -4.5]$	pass
Tagged, $K=2$ (H2)	bounded	256–724	256–1,024	max/min ≤ 8	pass (2.8)
Price only, $K=3$ (H3)	-10	-9.81	-9.96	slope in $[-12.5, -7.5]$	holds

Synthetic groups at rates $r_c+\Delta\eta\, s$ (s the K-point Gauss nodes; $r_c=0.6$ and $\Delta\eta$ from 0.5 to 0.125 for $K=2$, $r_c=0.5$ and $\Delta\eta$ from 0.6 to 0.25 for $K=3$), $a=0.5$, $\kappa=0.8$, total weight 0.08, $\sigma=10^{-3}$, $\sigma_f=0.02$. The test passes if H1 and H2 hold; H3 is reported either way. The tagged row gives the range of N^* over the spacing grid. Requirements above 4,096 use the central-limit draw described in the text. Requirements at each spacing are in Supplement Table 22. Test opened once.

Table 13: General memory: moment-matched cohort pairs and the moment estimator

A. Best approximation of $\lvert x\rvert$						
L	2	4	6	10	20	30
$L\,E_L(\lvert x\rvert)$	0.2500	0.2705	0.2756	0.2785	0.2797	0.2800
B. Pairs of Lemma 5.6						
Atoms per measure n	2	4	6	8	10	12
$W_1(\mu_A,\mu_B)$, $\rho=0.9$	0.2250	0.1037	0.0669	0.0494	0.0391	0.0324
$n\,W_1$, $\rho=0.9$	0.450	0.415	0.402	0.395	0.391	0.388
$\lVert k^A-k^B\rVert_2$, $\rho=0.9$	4.41×10^{-2}	1.03×10^{-3}	4.08×10^{-5}	1.96×10^{-6}	1.05×10^{-7}	5.95×10^{-9}
$\lVert k^A-k^B\rVert_2$, $\rho=0.99$	7.41×10^{-2}	5.84×10^{-3}	1.07×10^{-3}	2.76×10^{-4}	8.33×10^{-5}	2.76×10^{-5}
C. Linear-programming estimator, continuous memory distribution, $\rho=0.9$						
$\log_{10} N$	4	8	12	16	20	24
Moments used $(L+1)$	2	3	5	6	7	8
Error $d(\hat{\mu},\mu)$	0.00878	0.00759	0.00432	0.00445	0.00382	0.00399
Error $\times\log(N/\sigma^2)$	0.202	0.245	0.179	0.226	0.229	0.276

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\ldots,2n-2$ agree to 10^{-61} (60-digit arithmetic); every W_1 exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; L from the rule in Theorem 5.7(ii); measure fitted on a 401-point grid; 20 replications.

Table 14: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.62	0.0%
I_1: prices and signed flow	public flow	12	193.62	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.62	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.62	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

Table 15: Price-only round-trip probing loss per market and annualised weekly-renewal cost (hypothetical outside prober)

	Loss / probe (USDT)		Annual (52×, USDT)	
Economy ($\Delta\eta$)	Roll	1-sec	Roll	1-sec
$K = 2$, known (0.20)	6,200	149,901	322,400	7,794,852
$K = 2$, known (0.10)	99,192	2,398,422	5,157,984	124,717,944
$K = 2$, known (0.04)	3,874,696	93,688,369	201,484,192	4,871,795,188
$K = 2$, Radau (0.20)	4,292	103,788	223,184	5,396,976
$K = 2$, Radau (0.10)	283,488	6,854,612	14,741,376	356,439,824
$K = 3$, Radau (0.20)	804,403	19,450,107	41,828,956	1,011,405,564
$K = 3$, Radau (0.10)	1,137,325,880	27,500,017,407	59,140,945,760	1,430,000,905,164
$K = 4$, Radau (0.20)	156,632,086	3,787,291,895	8,144,868,472	196,939,178,540

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact K-cohort quadrature pair of Theorem 5.4 and Table 9 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem D.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning; it is a private transfer to other traders, not a social or resource cost, and it is not comparable with audit-trail budgets.

Table 16: Targeted versus market-wide intervention: two economies that prices barely separate

K	$\Delta\eta$	$\lvert S^A - S^B \rvert$	$\kappa q^G_{\max}$	$\kappa q^R_{\max}$	Window for $\bar{s}$	N for a correct choice
2	0.20	3.9×10^{-3}	0.112	0.144	[0.527, 0.562]	5
2	0.10	5.0×10^{-4}	0.093	0.131	[0.532, 0.570]	306
2	0.05	6.5×10^{-5}	0.086	0.125	[0.536, 0.575]	18,095
3	0.20	2.8×10^{-4}	0.073	0.092	[0.583, 0.603]	1,092
3	0.10	7.8×10^{-6}	0.071	0.076	[0.587, 0.592]	1.4×10^6

Calibration $a = 0.5$, $\kappa = 0.8$, $W = 0.08$, $r_c = 0.60$, observation noise $\sigma = 0.01$ per tap; G and R are the Gauss and Gauss–Radau economies. Both economies have $S \approx 0.66$–0.68. "Window" is the range of thresholds in which halting the largest group restores $S \le \bar{s}$ in one economy and not in the other. The last column is the exact Gaussian-shift requirement for a choice that is correct with probability 3/4 in both.

Table 17: Fitted persistence of trader-class positions, CFTC Commitments of Traders, eight regulated futures markets

Market	Weeks	Fast $\hat{r}$ [95% CI]	Slow $\hat{r}$	$\hat{r}_{\text{fast}} - \hat{r}_{\text{slow}}$ [95% CI]	$\hat{w}_{\text{fast}}$	Ratio
Bitcoin (CME)	434	0.951 [0.928, 0.969]	0.986	−0.035 [−0.062, −0.010]	−	23.7
Ether (CME)	282	0.890 [0.826, 0.932]	0.960	−0.070 [−0.143, −0.011]	−	9.1
E-mini S&P 500	1,039	0.914 [0.884, 0.937]	0.954	−0.040 [−0.071, −0.012]	−	18.3
10-year T-note	1,039	0.992 [0.985, 0.998]	0.993	−0.000 [−0.006, +0.007]	(0)	71.4
Euro	1,035	0.961 [0.947, 0.972]	0.996	−0.034 [−0.051, −0.022]	+	37.3
Yen	1,035	0.955 [0.939, 0.968]	0.984	−0.029 [−0.047, −0.013]	+	34.0
WTI crude oil	1,039	0.961 [0.946, 0.974]	0.999	−0.037 [−0.053, −0.023]	+	35.2
Natural gas	1,039	0.981 [0.969, 0.990]	0.980	+0.001 [−0.014, +0.017]	+	44.0

Pre-specified hypotheses: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass (two indistinguishable).

"Fast" (designated speculative) class: leveraged funds (financial futures) or managed money (commodities); "slow" (designated comparison) class: asset managers or producers/merchants. The labels are fixed by class role before estimation, not by the estimates; in natural gas the designated fast class has the (insignificantly) higher persistence. $\hat{w}_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0, 0.99]$ in every market. Weekly returns: Binance spot (Bitcoin, Ether); S&P 500 index, euro and yen spot, minus the change in the 10-year yield, and front-month crude-oil and natural-gas futures from Yahoo Finance. For the six non-futures series the spot–futures basis is ignored; a post hoc rerun with CME futures prices gives the same counts and the same signs of $\hat{w}_{\text{fast}}$ (see text). Test opened once.

Table 18: Fitted persistence rates with and without class-tagged data

Symbol	Days	Top position $\hat{r}$ [95% CI]	All accounts $\hat{r}$ [95% CI]	Width ratio	Classes differ
BTCUSDT	48	0.9889 [0.9842, 0.9935]	0.9986 [0.9955, 1.0005]	121.7	yes
ETHUSDT	48	0.9793 [0.9616, 0.9907]	0.9972 [0.9934, 0.9997]	55.6	yes
CELRUSDT	48	0.9884 [0.9830, 0.9922]	0.9882 [0.9804, 0.9914]	81.6	no
CVXUSDT	33	0.9822 [0.9732, 0.9908]	0.9829 [0.9694, 0.9900]	45.9	no
Pre-specified test on fresh days, 30 USD-M perpetuals, 2025–2026					
(i) width ratio > 10			30 of 30 (range 27.6–124.0)		
(ii) price-only rate interval wider than 0.25			30 of 30 (range 0.65–0.99)		
(iii) merged response $\sum_m k_m$ identified			5 of 30		
Classes differ			11 of 30		

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. Estimates are from day-demeaned regressions, which bias persistence down; corrected by a split-panel jackknife, the median tagged rate is 1.0008 (Section 6.5). The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification pre-specified before the fresh days were downloaded; a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

Table 19: Pre-specified out-of-sample cohort-state trajectory test on untouched 2nd-and-9th-of-month days, 2025–2026

30-min post-shock target (19 thin)	R^2_{pf}	R^2_{tag}	ΔR^2 (pts) [95% CI]	CW t (p)	Res.
Prim. 1: Divergence d_{t+6} (9,875 shocks)	−4.766	45.962	+50.728 [+45.291, +55.918]	21.27 ($< 10^{-16}$)	pass
Prim. 2: Top state $n_{\text{top},t+6}$ (9,875 shocks)	4.441	54.845	+50.404 [+47.226, +53.724]	21.85 ($< 10^{-16}$)	pass
Sec.: Divergence d_{t+6}, all (210,144 bars)	−0.586	37.053	+37.639 [+36.232, +39.107]	72.75 ($< 10^{-16}$)	—
Sec.: Top state $n_{\text{top},t+6}$, all (210,144 bars)	0.186	46.779	+46.593 [+45.555, +47.643]	80.44 ($< 10^{-16}$)	—

Out-of-sample R^2 in percent. Trained on 24 dates in 2024 (12,046 shocks, 120,384 bars); evaluated once on 42 untouched test dates (2nd and 9th of each month, Jan 2025–Sep 2026); 2,000-draw date-block bootstrap 95% CIs; Holm $p < 10^{-16}$ across both primaries.

Table 21: Probing protocols at observation horizon $T = 4$ per cycle

Protocol	δ	Executed vol.	Max trade	Error (SE)	Capital
A: single pulse (unconstrained)	0.03	131.5	131.5	0.047 (0.003)	6,985.91
	0.05	47.3	47.3	0.054 (0.004)	921.73
	0.08	18.5	18.5	0.053 (0.004)	146.73
	0.12	8.2	8.2	0.044 (0.003)	31.45
B: repeated capped pulses	0.03	2,768.8	6.2	0.050 (0.003)	7,035.75
	0.05	362.5	6.2	0.054 (0.004)	923.88
	0.08	56.2	6.2	0.047 (0.003)	146.00
	0.12	12.5	6.2	0.038 (0.003)	34.88
C: sequential test, capped	0.03	1,410.2	6.2	0.045 (0.003)	3,585.14
	0.05	195.6	6.2	0.038 (0.003)	500.00
	0.08	35.9	6.2	0.032 (0.003)	94.30
	0.12	10.5	6.2	0.011 (0.002)	29.85
D: generalized eigenvector	0.03	2,359.7	3.7	0.050 (0.003)	2,632.77
	0.05	314.8	3.7	0.048 (0.003)	353.67
	0.08	49.2	3.7	0.046 (0.003)	57.53
	0.12	9.9	3.8	0.045 (0.003)	13.56

2,000 replications per hypothesis and cell. Log–log capital slopes: -3.90, -3.84, -3.47, -3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D.

Table 22: Real-noise injection: number of experiments N^* at each spacing

	$\Delta\eta$	Price, real	Price, Gaussian	Tagged, real	Tagged, Gaussian
$K = 2$	0.500	64	64	724	1,024
	0.400	362	362	362	512
	0.320	1,448	1,024	362	362
	0.250	5,793	5,793	256	256
	0.200	16,384	23,170	256	362
	0.160	65,536	92,682	256	256
	0.125	370,728	370,728	256	256
$K = 3$	0.60	23,170	16,384		
	0.50	65,536	65,536		
	0.40	741,455	741,455		
	0.32	8,388,608	5,931,642		
	0.25	94,906,266	94,906,266		

Real noise: 30,219 one-minute log-price changes of Binance BTCUSDT on 21 days (the 6th of each month, January 2025 to September 2026), kurtosis 19.4. Tagged requirements were computed for $K = 2$ only.

Table 23: Numerical stress tests (numerical checks, not proofs)

Result	What was checked	Outcome
Upper bound, Theorem 5.5	Prony's requirement N^* for random separated configurations (15 for $K = 2$, 12 for $K = 3$); slope in $\Delta\eta$ and whether $N^*(\Delta\eta)^{4K-2}/\sigma^2$ grows as the spacing shrinks	Slopes -6.72 to -5.79 ($K = 2$, theory -6) and -11.01 to -9.29 ($K = 3$, theory -10); the normalized constant changes by a median factor of 1.05 and 0.94 from the widest to the narrowest spacing (at most 2.1); no configuration failed
Passive data, Theorem 5.9	Exact Whittle divergence rate between two moment-matched economies, 10 random settings for each K	Slopes 6.19–6.34, 10.27–10.40 and 14.33–14.54 for $K = 2, 3, 4$ (theory 6, 10, 14): the divergence shrinks at least as fast as the theorem requires
General memory, Lemma 5.6	Alternation pairs on $[0, \rho]$ for 12 random $\rho \in [0.3, 0.99]$ and $n = 2, \ldots, 6$: equal low moments, $W_1 \geq \rho E_{2n-2}$, kernel distance bound	60 of 60 cases pass
Value of disclosure, Theorem 6.2	Tagged and price-only requirements over the spacing grid, $K = 2, 3, 4$	Tagged requirement does not grow as the spacing shrinks (it varies by factors 2.8, 11.3 and 5.7 over the grid); price-only slopes -6.01 and -9.47 for $K = 2, 3$; for $K = 4$ our Prony implementation fails numerically at every spacing, so no price-only comparison

Gaussian observation noise, $\sigma = 10^{-3}$ (prices) and $\sigma_f = 0.02$ (tagged flow), $a = 0.5$, $\kappa = 0.8$, total weight 0.08, 200 replications per point, accuracy: every weight within 25% of the smallest true weight in at least 75% of replications. Moment-matched economies for the passive check: binomial weights on equally spaced nodes (the pair used in the Lean proof of Theorem 5.4). Seed 20261031.

Table 24: Account-level initial capital at $\delta = 0.05$ (par units)

Design	C_0	Error	End inv.	Peak margin	Max $\lvert v_t \rvert$	Trades
12 buys, $T = 20$ apart, never unwound	229.13	0.0578	75.00	39.38	6.25	12
6 buy/sell round trips	189.45	0.0308	0.00	3.28	6.25	12
5 buy/sell round trips	157.89	0.0442	0.00	3.28	6.25	10
12 buys, each unwound after 20 steps	378.82	0.0040	0.00	3.28	6.25	24
Generalized-eigenvector probe ($L = 120$)	65.06	0.0500	0.00	2.74	1.24	120
Generalized-eigenvector probe ($L = 160$)	64.86	0.0500	0.00	2.17	1.09	160
Fee-refined probe ($L = 240$)	64.48	0.0500	0.00	—	1.30	240

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The numerically evaluated lower bound of Theorem D.2 is 61.41; the explicit constant of Lemma 4.5 is 4.9×10^{-4}.

Table 25: Capital lower bounds and feasible upper bounds across separations (par units)

δ	ω^*	$C^*_{\mathrm{det,quad}}$	$\delta^4 C^*_{\mathrm{det,quad}}$	Certified C_{det}	Feasible UB	Adaptive LB
0.02	0.2546	2397.85	3.84×10^{-4}	2399.58	2428.25	1176.04
0.03	0.2541	473.19	3.83×10^{-4}	473.97	484.50	232.51
0.05	0.2526	61.14	3.82×10^{-4}	61.41	64.48	30.22
0.08	0.2488	9.26	3.79×10^{-4}	9.37	10.38	4.64
0.10	0.2453	3.77	3.77×10^{-4}	3.83	4.44	1.91
0.15	0.2331	0.73	3.67×10^{-4}	0.76	1.01	0.38

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. the numerical-bound column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \leq 240$; the adaptive column adds the fee lower bound to the adaptive bound.

Table 26: Finite-horizon capital at $\delta = 0.05$ (par units)

Horizon L	Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$	Full bound $C_{\mathrm{RH}}(L)$	Best feasible probe	Gap to $C_{\mathrm{det}} = 61.41$
80	62.55	64.72	65.02	5.9%
120	61.84	63.39	64.59	5.2%
160	61.54	62.46	64.50	5.0%
240	61.32	61.76	64.48	5.0%
320	61.25	—	≤ 64.48	$\leq 5.0\%$
480	61.19	—	≤ 64.48	$\leq 5.0\%$

Quadratic bound $c_{\mathrm{req}}/\lambda_{L-1}$ from the generalized eigenproblem of Theorem D.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with λ' computed against T_{h_0} rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (30) over 3,600 values of θ; at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

Table 27: Capital lower bound under alternative observation-noise calibrations (USDT)

Noise calibration	$\delta = 0.02$	$\delta = 0.05$	$\delta = 0.10$
Roll estimator, median (0.40 USDT)	3,874,696	99,192	6,200
One-second price-change s.d., median (1.98 USDT)	93,688,369	2,398,422	149,901
Earlier assumption (4 USDT, upper scenario)	383,812,500	9,825,600	614,100

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with σ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Table 28: Daily observation-noise estimates, Binance BTCUSDT one-second closes

Date	Roll $\hat{\sigma}$ (USDT)	One-second change s.d. (USDT)	Mean price (USDT)
2023-01-10	0.152	0.679	17,288
2023-01-11	0.092	0.891	17,456
2023-01-12	0.232	2.292	18,343
2023-01-13	0.348	1.762	19,075
2023-01-14	0.773	3.511	20,818
2023-01-15	0.482	1.790	20,778
2023-01-16	0.629	2.357	21,036
2023-01-17	0.457	1.976	21,194
2023-01-18	0.695	2.541	21,122
2023-01-19	0.402	1.316	20,850
2023-01-20	0.316	2.038	21,306
Median	0.402	1.976	

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

Table 29: Merged-measure equivalence and the δ^{-4} law under stylized nonlinear rules and closed-loop trained online learners

Rule	Test A raw ($O(A^2)$)	Test A norm. ($O(A)$)	Test B slope ($\|\Delta h\|_2$)	Test B slope (capital)	Capital at $\delta = 0.05$	A_{sat} / R^2_{exp}
Panel A. Stylized nonlinear and adaptive rules (fixed operating point)						
Linear benchmark	—	—	2.007	−4.001	61.42	—
1. Momentum (tanh limit)	2.015	1.015	2.007	−4.001	61.47	0.834
2. Threshold rule, averaged	2.043	1.043	2.007	−4.001	61.42	0.049
3. Adaptive gain (RLS)	2.064	1.064	2.007	−4.001	61.46	0.751
4. Online neural readout	2.011	1.011	2.007	−4.001	61.48	0.751
5. Tabular value, smoothed	2.039	1.039	2.007	−4.001	61.44	1.271
Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)						
6. Recursive least squares	3.000	2.000	2.008	−4.014	54.58	$R^2 = 0.993$
7. Two-layer network (SGD)	1.878	0.878	2.007	−4.012	65.19	$R^2 = 0.994$
8. Policy gradient on profit	3.000	2.000	2.004	−4.006	58.05	$R^2 = 0.995$
9. Tabular Q-learner (TD(0))	2.108	1.108	1.994	−3.973	186,821	$R^2 = 1.000$
Panel C. Where the trained-learner results change (range over the four learners)						
Independent training streams	rel. gap $\leq 10^{-5}$		0.13 to 0.92	−1.75 to −0.38	—	(c) fails
Learning during the probe	rel. gap $\leq 1.1 \times 10^{-3}$		1.99 to 2.01	−4.02 to −3.98	—	(c) holds

Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past τ only for $A > 0.139$.
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure 7). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most 10^{-5}. A raw Test A slope above 2 means the gap is of higher order than A^2. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, lower bound of Theorem D.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital.

Table 30: Samples

Sample	Coverage	Size	Source
Binance spot panel, 8 pairs	96 dates, 2021–2024	66.3M one-second bars	Binance archive
Binance spot fee-event windows, 8 pairs	±20 days around 2022-07-08 and 2023-03-22	640 pair-days	Binance archive
Binance USD-M perpetuals, 30 symbols	90 dates, 2023–2026	2,565 symbol-days	Binance archive
Binance BTCUSDT one-second panel	72 dates, 2021–2023	6.2M one-second bars	Binance archive
Binance BTCUSDT noise-calibration days	January 10–20, 2023	11 days	Binance archive

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table 31: Multi-year Binance spot panel: reduced-form indices under three timing constructions

Construction	$\hat{Q}$	Date interval	$\hat{S}$	Date interval	$\hat{Q} > 0$
Forward-paired flow (look-ahead)	0.0081424	[0.0072965, 0.0089882]	0.92074	[0.91665, 0.92482]	87.4%
Observed-flow EMA	0.0003733	[0.0002351, 0.0005116]	0.92072	[0.91640, 0.92504]	62.4%
Observed, 30 flow lags	0.0002580	[0.0001415, 0.0003745]	0.91738	[0.91167, 0.92308]	59.1%
Observed raw return	−0.0001555	[−0.0003732, 0.0000622]	−0.04051	[−0.05078, −0.03025]	49.2%
Placebo random walk	0.0000016	[−0.0000020, 0.0000052]	0.93510	[0.93485, 0.93534]	51.0%

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

Table 32: Asset means under forward-paired and observed flow

Pair	Forward-paired $\hat{Q}$	Observed $\hat{Q}$	$\hat{Q}_{30}$	Observed $\hat{S}$
BTC/USDT	0.0141870	0.0006073	0.0004891	0.93464
ETH/USDT	0.0130885	0.0005675	0.0002549	0.93511
SOL/USDT	0.0087763	0.0002129	0.0001100	0.91966
ADA/USDT	0.0067465	0.0003928	0.0003546	0.91452
DOGE/USDT	0.0058956	0.0002464	0.0001432	0.92496
BNB/USDT	0.0057420	0.0004479	0.0002762	0.89533
XRP/USDT	0.0057034	0.0002249	0.0001508	0.92362
LTC/USDT	0.0049997	0.0002870	0.0002853	0.91792

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table 33: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

	Real prices	Placebo (independent random walk)
Mean loop gain $\hat{Q}$	0.0045	0.0000009
95% CI (day bootstrap)	[0.0034, 0.0056]	[−0.0000006, 0.0000023]
Mean price persistence $\hat{J}_{11}$	0.9364	0.9357
Mechanical value $1 - 2/31$		0.9355

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table 31.

Table 34: Feedback estimators under order splitting (simulation, 500 replications)

Estimator	Coverage at $\gamma = 0.2$	Rejection at $\gamma = 0$	Bias	RMSE
OLS	0.000	1.000	0.892	0.893
Cochrane–Orcutt, true ρ	0.944	0.054	0.001	0.039
Cochrane–Orcutt, estimated ρ	0.916	0.080	0.002	0.043
2SLS	0.938	0.044	−0.002	0.116
LIML	0.938	0.044	−0.003	0.117

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future z) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

Table 35: Zero fees on BTCUSDT: activity and measured feedback

	Zero fees introduced (2022-07-08)	Zero fees removed (2023-03-22)	Pooled
Log quote volume, $\hat{\beta}$	0.945 (0.074)	1.599 (0.080)	1.272 (0.084)
Observed-flow $\hat{Q}$, $\hat{\beta}$	0.00058 (0.00032)	−0.00091 (0.00060)	−0.00016 (0.00036)
p-value	0.078	0.140	0.650
Pair-days	320	320	640

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.